

BrokerVerse OOTB

Insurance broking platform by iorta TechNXT

Solution Architecture · Document 02 of 11

Database Design

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Version	Date	Author	Change
1.0	29 September 2026	iorta TechNXT	Initial issue
1.1	29 September 2026	iorta TechNXT	Updated to the current system: placement journey, co-insurance, posting rules, period end and BIR tax, bank reconciliation, document numbering, commission rate matrix, single PDF engine, uploads and go-live imports, housekeeping, business time zone, broker roles, deployment on the BrokerVerse URL
1.2	04 October 2026	iorta TechNXT	New modules (AML/CFT, BIR, payables and fixed assets, distribution, fleet and marine, My Work, go-live workbench), integration framework, encryption and masking of personal data, branding and e-signatures, release pipeline; database at migration 0322 (reference data), new ER diagrams

Review and approval

Role	Name	Signature	Date
Solution architect			
DevOps / infrastructure lead			
Information security			
Business owner			

Conventions

- Statements of fact are derived from the code, configuration and database of the baseline above (file paths are given relative to the repository). Figures measured on the local test system are labelled **test data** or **indicative**.
- Targets, sizes, retention periods and thresholds that are not facts of the code are labelled **Recommended / to be confirmed by the business and DevOps**.
- Items marked **Gap** are defects or missing capabilities found while preparing this document.

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1 Introduction

1.1 Purpose and scope

This document describes the logical and physical design of the BrokerVerse OOTB database: the schema by business domain, the entity-relationship structure of each domain, the keys, the JSONB document columns, document numbering, audit and soft-delete conventions, the posting rules and period controls that protect the ledger, the migration and seed approach, and the handling of time zones, dates and amounts. The table-by-table inventory with sizes, row counts and retention classes is document 03.

Sources: `backend/src/db/migrations/*.sql` (139 files, 8,360 lines, plus the 10 migrations of packages B and G being merged), `backend/src/db/seeds`, `backend/src/db/pool.js`, `migrate.js`, `seed.js`, `backend/src/lib/dates.js`, `money.js`, `numbering.js`, and the catalogue of a PostgreSQL 16 database created for this issue on 04 October 2026: migrated with the migrations of the branch and of packages B and G and seeded with the reference data only (snapshot in `docs/architecture/tools/data/db_snapshot.json`).

1.2 Database platform

Item	Value (local test database)
PostgreSQL version	16.13 (Ubuntu 16.13-0ubuntu0.24.04.1)
Database size	28.6 MB (test data)
Server time zone	Asia/Manila
Tables (schema public)	260
Columns by type	text 2595, timestamp with time zone 590, numeric 493, integer 293, date 220, jsonb 166, bigint 116, boolean 93, ARRAY 22, time without time zone 1
Indexes	901 (440 unique incl. 260 primary keys, 71 partial, 1 GIN)
Foreign keys	468 (60 without a supporting index)
Identity sequences (serial / identity columns)	125
Document-number counters (table sequences)	0 counters for -
Extensions	pgcrypto 1.3, plpgsql 1.0
Application functions	accounting_period_lock_guard, backfill_risk_participants, cities_default_region, document_numbering_mirror_prefix, format_document_number, gl_accounts_defaults, jv_check_posting, next_document_number, next_number, numbering_business_date, numbering_fiscal_year, numbering_optional_token, numbering_period_key, numbering_prefix, numbering_start_number, pe_balance_before, ph_place_key, pii_blind_index, pii_decrypt, pii_encrypt, pii_json_key, pii_jsonb_needs, pii_protect_columns, pii_protect_jsonb, pii_setting, pii_store, policies_channel_inherit, quotes_channel_inherit, quotes_link_product, receipt_applications_collected_on, rp_insurer_id, rp_parts_from_doc, rp_share, rp_write_split, rpt_age_bucket, set_updated_at, user_effective_roles

Item	Value (local test database)
Triggers	accounting_period_lock_guard_trg (accounting_periods), audit_log_pii (audit_log), bir_2307_certificates_pii (bir_2307_certificates), cities_region (cities), clients_pii (clients), commission_rates_updated (commission_rates), commission_referrers_pii (commission_referrers), document_numbering_mirror (document_numbering), document_numbering_updated (document_numbering), gl_accounts_defaults_trg (gl_accounts), insurer_rate_tables_updated (insurer_rate_tables), jv_check_posting_trg (journal_vouchers), leads_pii (leads), lgu_tax_rates_updated (lgu_tax_rates), package_bundles_updated (package_bundles), package_quotes_updated (package_quotes), package_sections_updated (package_sections), payment_gateways_updated (payment_gateways), payment_links_updated (payment_links), placements_pii (placements), policies_channel_inherit (policies), policies_pii (policies), premium_charge_rules_updated (premium_charge_rules), quotes_channel_inherit (quotes), quotes_link_product (quotes), quotes_pii (quotes), receipt_applications_collected_on (receipt_applications), regions_updated (regions), roles_updated (roles), users_updated (users)
Views / materialised views	bank_account_links, bank_book_lines
Migrations applied	149 (0001_core.sql to 0322_quote_covers_and_risk_fields.sql)
Application settings (app_settings)	659 keys in 64 groups
Scheduled jobs (scheduled_jobs)	35
Snapshot taken	2026-10-04T09:13:21+00:00

One database and one schema (**public**) hold all data. The application connects with a single login through **DATABASE_URL** (node-postgres pool, maximum 10 connections per API process). The login needs the right to create tables because the API applies its own migrations on start. The database time zone is set to **Asia/Manila** (**ALTER DATABASE ... SET timezone**, or the RDS parameter group), as **deploy/README.md** requires. Only the **pgcrypto** extension is used, for **gen_random_bytes()** in primary-key defaults.

2 Schema overview by domain

The 260 tables fall into 25 domains (250 tables without packages B and G). Row counts and sizes are those of the reference data a new database starts with; transaction tables are empty.

Domain	Tables	Rows (reference data)	Size KB	Tables
Security and identity	15	434	736	access_review_items, access_reviews, authority_limits, authority_transaction_types, login_history, password_history, password_resets, permissions, refresh_tokens, role_permissions, roles, sod_rules, user_delegations, user_roles, users
Platform services	16	1,461	1,440	agent_events, app_settings, audit_log, document_numbering, document_signature_slots, documents, e_signatures, email_outbox, job_queue, job_runs, master_records, master_types, notifications, scheduled_jobs, schema_migrations, sequences
Reference masters	17	5,475	2,136	banks, branches, cities, countries, coverages, currencies, districts, insurance_companies, policy_types, postal_codes, products, regions, signatories, states, vehicle_brands, vehicle_models, vehicle_variants
Product configurator	9	91	496	insurer_rate_tables, lgu_tax_rates, package_bundle_sections, package_bundles, premium_charge_rules, product_components, product_risk_mappings, product_risk_sections, product_templates
Sales and policy	14	3	936	clients, cover_notes, endorsements, leads, package_endorsements, package_quotes, package_sections, payment_events, payment_gateways, payment_links, policies, policy_payments, quote_customer_responses, quotes
Placement and co-insurance	4	0	264	broker_slips, insurer_offers, placements, risk_participants
Claims	6	0	224	claim_document_items, claim_document_reminders, claim_field_changes, claim_history, claim_settlement_movements, claims
Renewals	7	0	224	renewal_activities, renewal_batch_policies, renewal_batches, renewal_notices, renewal_quotes, renewals, winback_campaigns
Billing, receipts and collections	15	4	648	client_credit_exceptions, collection_actions, collection_items, post_dated_cheques, premium_instalment_plans, premium_instalments, premium_warranty_actions, premium_warranty_extensions, receipt_applications, receipt_lines, receipts, receivable_credits, receivable_participants, receivables, write_off_reasons
General ledger and posting rules	8	361	512	accounting_config_changes, accounting_periods, entry_matches, gl_accounts, journal_lines, journal_vouchers, posting_rule_lines, posting_rules
Period end and tax	12	39	392	bir_2307_certificates, fiscal_years, opening_balances, period_close_checklist, period_close_entries, period_close_run_checks, period_close_runs, period_status_history, recurring_journal_runs, recurring_journals, tax_codes, year_end_runs
Bank reconciliation	9	17	304	bank_match_rules, bank_rec_match_items, bank_rec_matches, bank_reconciliation_history, bank_reconciliations, bank_statement_formats, bank_statement_lines, bank_statements, bank_transaction_types
Disbursements and petty cash	9	0	408	checkbooks, disbursements, invoice_lists, petty_cash_disbursements, petty_cash_funds, petty_cash_receipts, petty_cash_replenishments, petty_cash_request_lines, petty_cash_requests
Commission	8	0	272	commission_adjustments, commission_rates, commission_referrers, commissions, override_agreement_tiers, override_agreements, override_computations, override_settlements

Domain	Tables	Rows (reference data)	Size KB	Tables
Remittance and direct bill	16	1	648	commission_debit_note_collections, commission_debit_note_lines, commission_debit_notes, direct_bill_client_payments, direct_bill_items, insurer_refund_credits, insurer_statement_formats, insurer_statement_lines, insurer_statement_resolutions, insurer_statements, remittance_allocations, remittance_approvals, remittance_delegations, remittance_items, remittance_lines, remittances
Reinsurance	10	0	376	cessions, fac_placement_shares, fac_placements, fac_settlements, reinsurance_bordereaux, reinsurance_exceptions, reinsurance_reconciliations, reinsurance_recoveries, reinsurance_treaties, reinsurers
Incentive	3	0	80	incentive_calculations, incentive_programs, incentive_results
Reporting	5	40	240	bi_extract_runs, generated_reports, report_builder_reports, report_definitions, report_schedules
Distribution and marketing	13	0	520	campaign_recipients, campaign_segments, campaign_templates, campaigns, channel_billing_accounts, comparison_reports, dealer_sales, dealer_sales_batches, distribution_channels, lead_assignment_history, lead_assignment_rules, motor_programmes, sales_activities
Fleet, marine and motor claims	8	0	344	claim_loas, claim_repair_estimates, claim_vehicle_releases, fleet_schedules, fleet_vehicles, open_cover_certificates, open_cover_declarations, open_covers
Payables and fixed assets	7	0	248	fixed_asset_depreciation, fixed_asset_disposals, fixed_assets, supplier_invoice_lines, supplier_invoices, supplier_payment_allocations, supplier_payments
BIR returns and invoicing	6	0	184	bir_return_filings, cas_book_prints, eis_submissions, sales_invoice_lines, sales_invoice_payments, sales_invoices
Compliance: AML, Insurance Commission and data privacy	27	80	1,040	aml_alerts, aml_cases, aml_edd_reviews, aml_list_entries, aml_list_versions, aml_provider_requests, aml_report_items, aml_reports, aml_risk_assessments, aml_risk_factors, aml_rules, aml_screening_hits, aml_screening_lists, aml_screenings, client_beneficial_owners, client_kyc_documents, client_signatories, complaint_reminders, complaints, compliance_fit_proper, compliance_licence_reminders, compliance_licences, data_subject_requests, ic_statement_lines, personal_data_breach_reminders, personal_data_breaches, privacy_consent
Integrations	12	19	416	bank_file_layouts, bank_payment_batch_lines, bank_payment_batches, coc_series, ctpL_authentications, insurer_api_mappings, integration_attempts, integration_connectors, integration_inbox, integration_outbox, message_templates, payee_bank_accounts
Go-live data load and work management	4	0	152	data_load_batches, data_load_comparisons, data_load_rows, work_tasks
Total	260	8,025	13,240	

3 Design conventions

3.1 Keys

Convention	Detail
Business records	103 tables use a text primary key made of a readable prefix and random hexadecimal characters generated by the database, for example <code>policies.id</code> <code>DEFAULT 'pol_' encode(gen_random_bytes(8), 'hex')</code> . Prefixes include <code>ld_</code> lead, <code>cl_</code> client, <code>qt_</code> quotation, <code>bs_</code> broker slip, <code>ofr_</code> insurer offer, <code>plc_</code> placement, <code>pol_</code> policy, <code>end_</code> endorsement, <code>clm_</code> claim, <code>rnw_</code> renewal, <code>or_</code> receipt, <code>jv_</code> journal voucher, <code>pv_</code> payment voucher, <code>rm_</code> remittance, <code>dn_</code> debit note, <code>brc_</code> bank reconciliation, <code>yec_</code> year-end run, <code>usr_</code> user and <code>doc_</code> document. The ids cannot be guessed and do not collide across tables.
Reference, line and log tables	125 tables use integer or bigint serial keys: masters (countries, banks, insurers, products), lines and logs that grow fast (<code>journal_lines</code> , <code>audit_log</code> , <code>login_history</code> , <code>job_runs</code> , <code>claim_history</code> , <code>risk_participants</code> , <code>bank_rec_match_items</code> , <code>period_status_history</code>).
Natural and composite keys	32 tables: <code>app_settings(key)</code> , <code>gl_accounts(code)</code> , <code>master_types(code)</code> , <code>report_definitions(code)</code> , <code>accounting_periods(period)</code> , <code>fiscal_years(code)</code> , <code>tax_codes(code)</code> , <code>bank_statement_formats(code)</code> , <code>bank_transaction_types(code)</code> , <code>bank_match_rules(code)</code> , <code>period_close_checklist(code)</code> , <code>opening_balances(fiscal_year, account_code)</code> , <code>receivable_participants(receivable_id, insurance_company_id)</code> , <code>refresh_tokens(jti)</code> , <code>schema_migrations(name)</code> , <code>sequences(name, period)</code> , <code>role_permissions</code> , <code>user_roles</code> and others.
Business numbers	Readable document numbers (<code>QT-2026-00001</code> , <code>POL-2026-00001</code> , <code>OR-2026-00019</code> , <code>BS-</code> , <code>PS-</code> , <code>MEC-</code> , <code>BRC-</code>) are separate unique columns (<code>quote_number</code> , <code>policy_number</code> , <code>receipt_number</code> , <code>slip_number</code> , <code>run_number...</code>), never primary keys. Most detail endpoints accept either the id or the number.
Foreign keys	468 foreign keys: 369 <code>NO ACTION</code> (a referenced record cannot be deleted while in use), 77 <code>CASCADE</code> (children with no meaning without their parent: journal lines, receipt lines, match items, claim history, renewal notices, role grants, refresh tokens, notifications) and 22 <code>SET NULL</code> .

3.2 Column types

Data	Type and handling
Amounts	<code>numeric</code> (490 columns), typically <code>numeric(14,2)</code> ; converted to JavaScript numbers by the pool type parser; all rounding through <code>round2()</code> in <code>lib/money.js</code> (half away from zero, the same rule as PostgreSQL <code>round(numeric, 2)</code>). Shares are <code>numeric(7,4)</code> .
Business dates	<code>date</code> (217 columns: inception, expiry, due, loss, receipt, voucher, statement and period dates), returned to the application as <code>YYYY-MM-DD</code> strings so no time-zone shift can occur. Date columns that default to today take <code>numbering_business_date()</code> , the date in <code>general.timezone</code> .
Instants	<code>timestamptz</code> (590 columns, for example <code>created_at</code> , <code>approved_at</code> , <code>sent_at</code>); there are no <code>timestamp without time zone</code> columns.
Flexible parts	<code>jsonb</code> (166 columns in 103 tables), see below.
Codes and statuses	<code>text</code> with validation in the application (zod, service rules) and 390 <code>CHECK</code> constraints; statuses are lower-case codes mapped to display labels.

3.3 JSONB document columns

JSONB is used where the structure varies by product, screen or configuration. Columns that are searched, joined, totalled or reported on stay relational.

Table	JSONB columns and use
quotes, broker_slips, placements	Wizard inputs and selected covers, the full document (doc) with premium breakdown and customer data, risk details and requested covers
policies	<code>details</code> (risk and party details, renewal link), <code>doc</code> (policy document data carried from the quotation or placement)
insurer_offers	<code>taxes</code> , <code>deductibles</code> , <code>terms</code> of an insurer's answer
claims	<code>details</code> , <code>driver</code> , <code>third_party</code> , <code>policy_info</code> , <code>adjuster</code> , <code>settlement</code>
endorsements	<code>changes</code> (field changes and re-pricing), <code>endorsement_type_ids</code> , <code>completion</code>
renewals, renewal_quotes	Coverage details, accessories, order summary, limits, premium breakdown, taxes and rating
product_templates, product_components	<code>config</code> , <code>features</code> , <code>insurers</code> , <code>tags</code> ; component <code>data</code> (covers, rating tables, rules, the motor CTPL tariff)
master_records, master_types	<code>data</code> of generic masters (GIN index <code>master_records_data_idx</code> , <code>jsonb_path_ops</code>); <code>fields</code> and <code>unique_keys</code> of each master screen
app_settings	<code>value</code> : every configuration value
audit_log	<code>before_data</code> , <code>after_data</code> of every audited change
period end and bank reconciliation	<code>period_close_runs.steps</code> , <code>period_close_run_checks.detail</code> , <code>recurring_journals.lines</code> , <code>year_end_runs.checks</code> , <code>bir_2307_certificates.lines</code> , <code>bank_statement_formats.columns</code> , <code>bank_match_rules.params</code> , <code>bank_reconciliations.snapshot</code>
finance and reinsurance	<code>remittances.data</code> , <code>remittance_items.data</code> , <code>remittance_approvals.history</code> , <code>insurer_refund_credits.split</code> , <code>reinsurance_treaties.terms</code> , <code>incentive_programs.structure</code>
operations	<code>scheduled_jobs.params</code> , <code>job_runs.output</code> , <code>job_queue.payload</code> , <code>generated_reports.params</code> , <code>report_definitions.parameters</code>

3.4 Document numbering

Every document number comes from a series in the Document Numbering master (`document_numbering`, 85 series, Master > Document Numbering). `nextDocumentNumber(code, { db })` in `lib/numbering.js` calls the SQL function `next_document_number(code, branch, lob, date)`. The function locks the counter row of the series and period in `sequences` until the caller's transaction ends, so two users never get the same number and a rolled-back save gives its number back. `format_document_number()` applies the pattern: tokens `{PREFIX}` `{YYYY}` `{YY}` `{MM}` `{FY}` `{BRANCH}` `{LOB}` `{SEQ}`; all seeded series use `{PREFIX}-{YYYY}-{SEQ}`. A prefix must be unique among active series (index `document_numbering_active_prefix`); migration 0246 lets a series start a period at the old system's next number at go-live. The reset rule of a series is `yearly`, `fiscal_yearly`, `monthly` or `never`; the counter period is `2026`, `FY2027`, `2026-09` or `ALL`. The year is taken from the business date in `general.timezone`, so a document created at 00:30 Manila time on 1 January gets the new year. The next number of a series can only be moved forward (for example to continue from the legacy system at go-live). A trigger copies each prefix into the read-only setting `numbering.<code>.prefix`.

Series (code)	Prefix	Reset	Series (code)	Prefix	Reset
acknowledgement_receipt	AR	yearly	marine_certificate	MIC	yearly
adjustment	ADJ	yearly	marine_declaration	MDC	yearly
amL_alert	AMA	yearly	open_cover	MOC	yearly

Series (code)	Prefix	Reset	Series (code)	Prefix	Reset
amL_case	AMC	yearly	override_computation	OVC	yearly
amL_edd	EDD	yearly	package_policy	PKG	yearly
amL_report	AMR	yearly	package_quote	PQ	yearly
bank_payment_batch	BPB	yearly	payment_link	PL	yearly
bank_reconciliation	BRC	yearly	pdv	PDC	yearly
bank_statement	BST	yearly	period_close	MEC	yearly
bank_txn	BNK	yearly	petty_cash	PC	yearly
bir_2307	CWT	yearly	petty_cash_fund	PCF	never
bordereau	BDX	yearly	petty_cash_receipt	PCRC	yearly
broker_slip	BS	yearly	petty_cash_request	PCR	yearly
campaign	WB	yearly	placement	PS	yearly
cession	CES	yearly	policy	POL	yearly
claim	CLM	yearly	product_template	TPL	yearly
claim_loa	LOA	yearly	quote	QT	yearly
claim_payment_voucher	CPV	yearly	receipt	OR	yearly
client	CL	yearly	receipt_txn	RT	yearly
commission_debit_note	DN	yearly	recurring_journal	RJV	yearly
commission_master	COM	never	reinsurer	RE	yearly
comparison_report	CMP	yearly	remittance	REM	yearly
complaint	CPX	yearly	remittance_batch	BLK	yearly
cover_note	CVN	yearly	remittance_bill	BIL	yearly
data_breach	PDB	yearly	remittance_exception	EXC	yearly
data_subject_request	DSR	yearly	remittance_notice	NTF	yearly
dealer_sales_batch	DSB	yearly	remittance_report	RPT	yearly
disbursement_txn	DT	yearly	remittance_schedule	SCH	yearly
dn_collection	DNC	yearly	renewal	RN	yearly
employee	EMP	never	renewal_batch	RB	yearly
endorsement	END	yearly	renewal_quote	RQ	yearly
fac_slip	FAC	yearly	ri_reconciliation	REC	yearly
fixed_asset	FA	yearly	ri_recovery	RCL	yearly
fleet_schedule	FLT	yearly	sales_invoice	SI	never
incentive_calc	CALC	yearly	settlement	SET	yearly
incentive_program	INC	yearly	statement	STMT	yearly
insurer_offer	OFR	yearly	supplier_invoice	APV	yearly
insurer_statement	ISR	yearly	supplier_payment	SPV	yearly
invoice	INV	yearly	transfer	TRF	yearly
invoice_list	IL	yearly	treaty	TRT	yearly
invoice_payment	PAR	yearly	voucher	PV	yearly
journal	JV	yearly	year_end_close	YEC	yearly
lead	LD	yearly			

3.5 Audit, history and soft delete

Convention	Detail
Row stamps	<code>created_at</code> (192 tables), <code>updated_at</code> (148), <code>created_by</code> (157), <code>updated_by</code> (107). <code>updated_at</code> is set by the application; triggers maintain it on <code>users</code> , <code>roles</code> , <code>commission_rates</code> and <code>document_numbering</code> (<code>set_updated_at()</code>).
Audit log	<code>audit_log(id, at, user_id, username, entity, entity_id, action, before_data, after_data, ip, source)</code> (<code>source</code> , migration 0247: screen, API or job), written by <code>lib/audit.js</code> from every create, update, approval, posting, run and sign-in related event; indexed on (<code>entity, entity_id</code>), (<code>at</code>) and (<code>username</code>).
Domain histories	<code>claim_history</code> , <code>claim_field_changes</code> , <code>renewal_activities</code> , <code>renewal_notices</code> , <code>collection_actions</code> , <code>remittance_approvals.history</code> , <code>period_status_history</code> , <code>bank_reconciliation_history</code> , <code>login_history</code> , <code>password_history</code> , <code>job_runs</code> .
Soft delete	<code>leads</code> , <code>quotes</code> and <code>bank_statements</code> carry <code>deleted_at</code> ; masters and product components use a <code>status</code> of <code>deleted</code> (unique indexes are partial: <code>WHERE status <> 'deleted'</code>). Financial documents are never deleted but cancelled or reversed (<code>receipts.cancelled_at</code> , journal reversal and correction vouchers, <code>bir_2307_certificates.cancelled_at</code>). A draft quotation is the only business record deleted physically.

3.6 Ledger integrity

Control	Detail
Posting rules	Every system journal is built by <code>postEvent()</code> from the posting rule of its business event (43 events with an active rule, for example <code>policy.issue.broker_billed</code> , <code>ap.invoice</code> , <code>fa.depreciation</code> , <code>sales_invoice.issue</code> , <code>override_commission</code> , <code>accrual</code> , <code>receipt.apply</code> , <code>remittance.settlement</code> , <code>claim.funds_received</code> , <code>commission.clawback</code>). A rule has versions with an effective date (<code>posting_rules</code>) and lines (<code>posting_rule_lines</code>) that name either a GL account or an account role. Account roles resolve to GL accounts through the settings <code>accounting.account.<role></code> (Account Determination). A line marked per participant is split across the co-insurers by share, with the rounding remainder on the lead.
Balance and period trigger	Trigger <code>jv_check_posting</code> refuses to post a journal voucher with fewer than two lines, unbalanced, with a zero total, or in a period that does not accept postings.
Period status	<code>accounting_periods</code> holds periods 1 to 12 and the adjustment period 13 of each fiscal year with the status <code>open</code> , <code>soft_closed</code> , <code>closed</code> or <code>locked</code> . Trigger <code>accounting_period_lock_guard</code> refuses to reopen a period locked by the year-end close. Every change is written to <code>period_status_history</code> .
Opening balances	The year-end close writes the balance-sheet balances into <code>opening_balances</code> of the next year; the go-live import writes the old system's trial balance there (source <code>go-live:<date></code>). Reports read them through <code>pe_balance_before()</code> ; no opening journal is posted.
Co-insurance	<code>risk_participants</code> holds the participants of a broker slip, quotation, placement or policy (one lead, shares total 100%, index <code>risk_participants_one_lead</code>); <code>receivable_participants</code> and <code>remittance_allocations</code> split bills and collections per insurer.
Account defaults	Trigger <code>gl_accounts_defaults</code> derives the normal balance of an account from its type.

3.7 Constraints and indexes

The schema has 901 indexes: 260 primary keys, 180 further unique indexes (business numbers, natural keys and partial uniqueness such as one open renewal per policy, one pending approval per entity, one active close run per period, one lead co-insurer per record), 71 partial indexes and one GIN index. Case-insensitive uniqueness uses expression indexes on `lower(...)`.

Migration `0146_foreign_key_indexes.sql` indexed the foreign keys of every table that grows (transactions, documents, lines and logs) and added the operational log indexes `job_runs` (`job_id`, `started_at`), `email_outbox` (`status`, `created_at`), `audit_log` (`at`) and `audit_log` (`username`). The list was produced by `backend/scripts/fk-index-report.js`. 60 of the 468 foreign keys have no index whose first column is the foreign-key column; they are on small master and set-up tables (vehicle models, states, policy types, product components) or are maker-checker stamp columns (`approved_by`, `rejected_by`), left out on purpose. They are listed in document 03.

4 Entity-relationship diagrams

The diagrams are generated from the database catalogue. Each box lists the primary key (PK), up to two unique keys (UK), the foreign keys (FK; a red asterisk marks an FK without a supporting index) and the JSONB columns ({ }). Dashed boxes are tables of another domain. References to [users](#) (creator, owner, approver) are listed as columns but drawn only in the security diagram.

4.1 Sales and policy

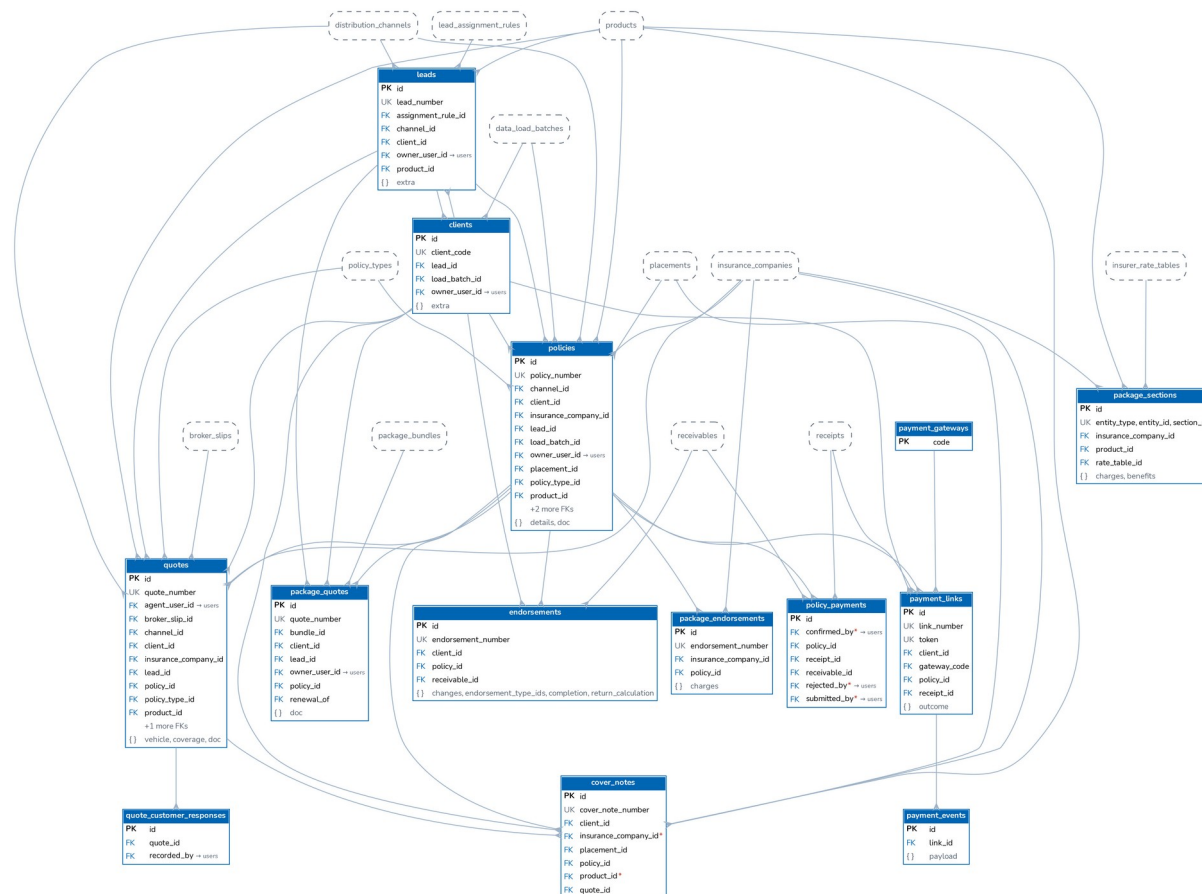


Figure 1. Sales and policy: leads, clients, quotations, policies, endorsements, payment captures

A lead becomes a client when a quotation is converted (**clients.lead_id** is unique when set). A quotation references the lead or client, product, policy type, insurer and signatory; conversion creates the policy (**policies.quote_id**, **quotes.policy_id**). A renewed policy points to its predecessor (**policies.renewed_from**, **renewed_to**). Endorsements belong to a policy and may raise their own bill (**endorsements.receivable_id**). **policy_payments** holds payments captured by front-office staff until Accounting confirms them and issues a receipt.

4.2 Placement and co-insurance

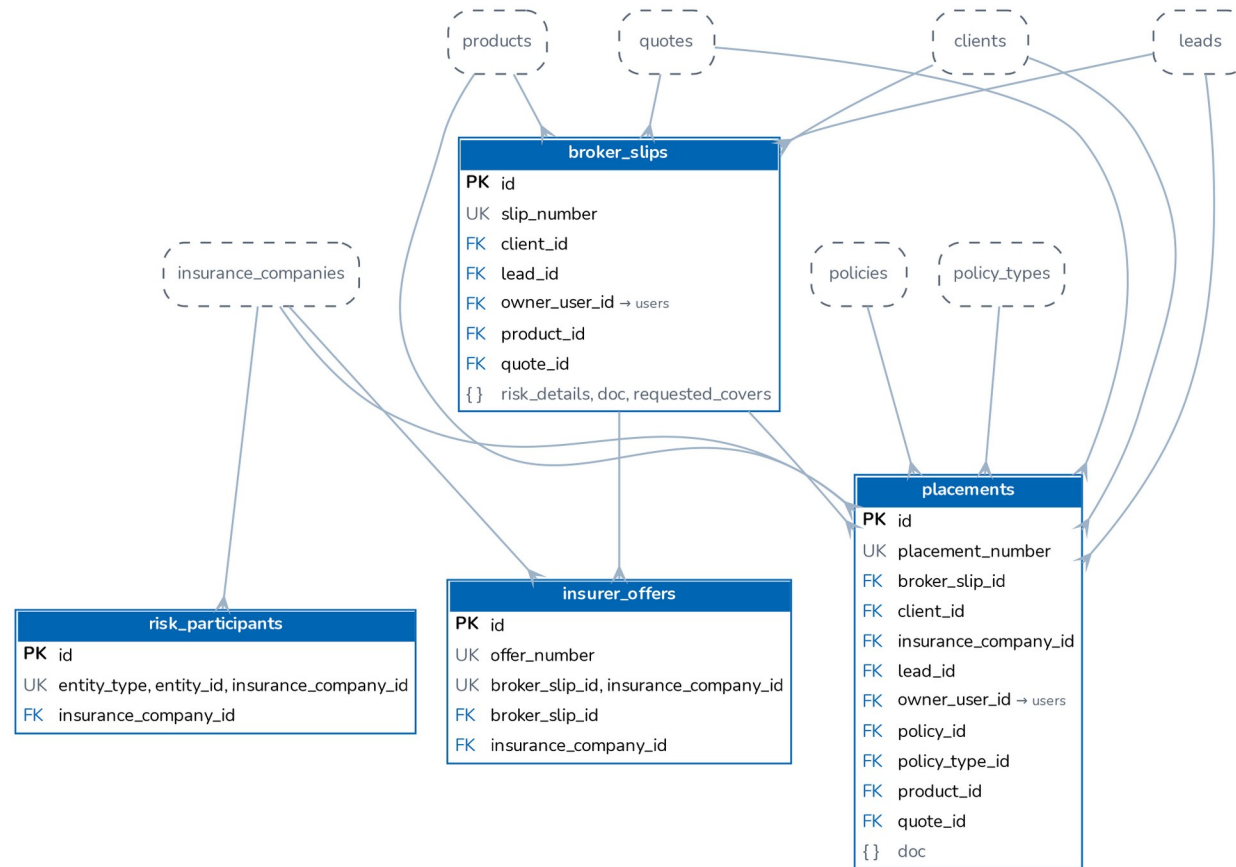


Figure 2. Placement: broker slips, insurer offers, placement slips and co-insurance participants

A broker slip is created from a lead or client and sent to several insurers; each answer is an **insurer_offers** row (premium, rate, taxes, deductibles, validity, share offered) or a decline. The chosen offers become a quotation or a placement slip (**placements.broker_slip_id**, **placements.quote_id**). A placement is sent to the lead insurer and the co-insurers; each participant binds with its own reference and the policy is issued when all have bound (**placements.policy_id**). **risk_participants** is shared by all four record types through **entity_type** and **entity_id**.

4.3 Renewals

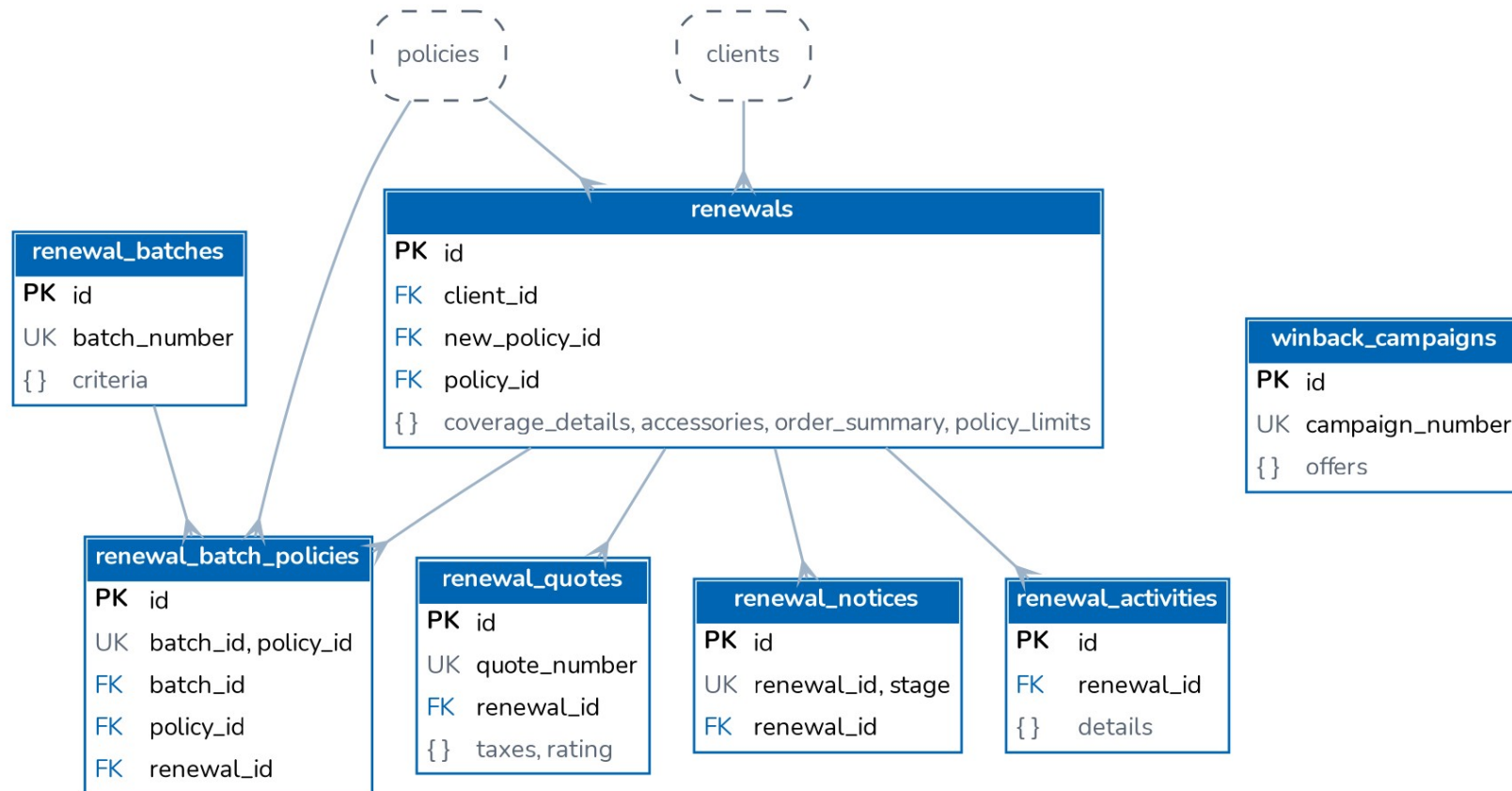


Figure 3. Renewals: renewal cases, quotes, notices, activities, batches, win-back campaigns

4.4 Billing, receipts and collections

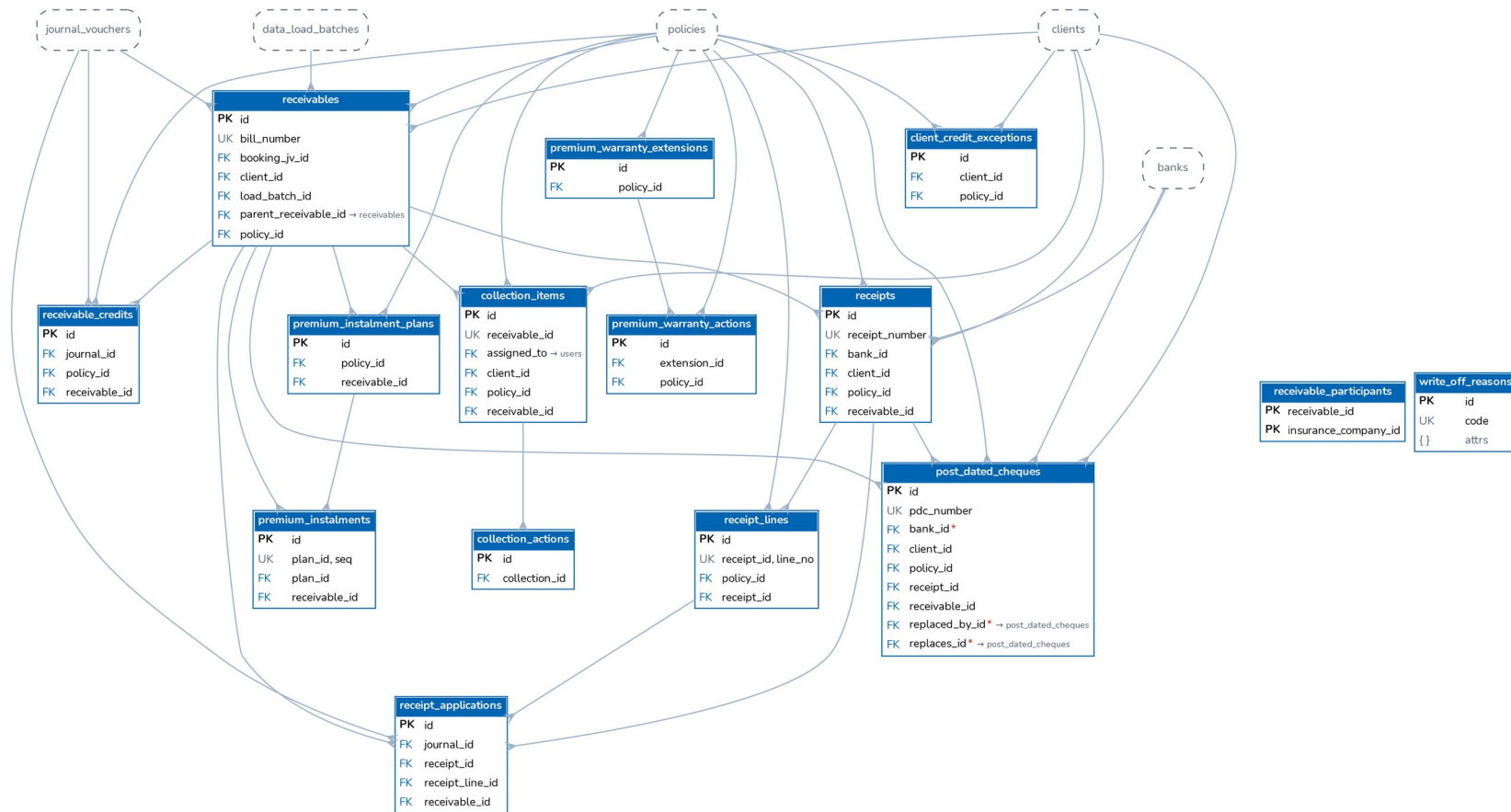


Figure 4. Billing, receipts, credits, co-insurance split and collections

A policy, endorsement or renewal raises a receivable (bill) with its booking journal (`receivables.booking_jv_id`). A receipt has lines per policy, and receipt applications allocate the lines to bills and link the posting journal. Return premium and cancellations reduce a bill through `receivable_credits`. Collection items track the follow-up of open bills; write-offs use the reasons and GL accounts of `write_off_reasons`.

4.5 General ledger and posting rules

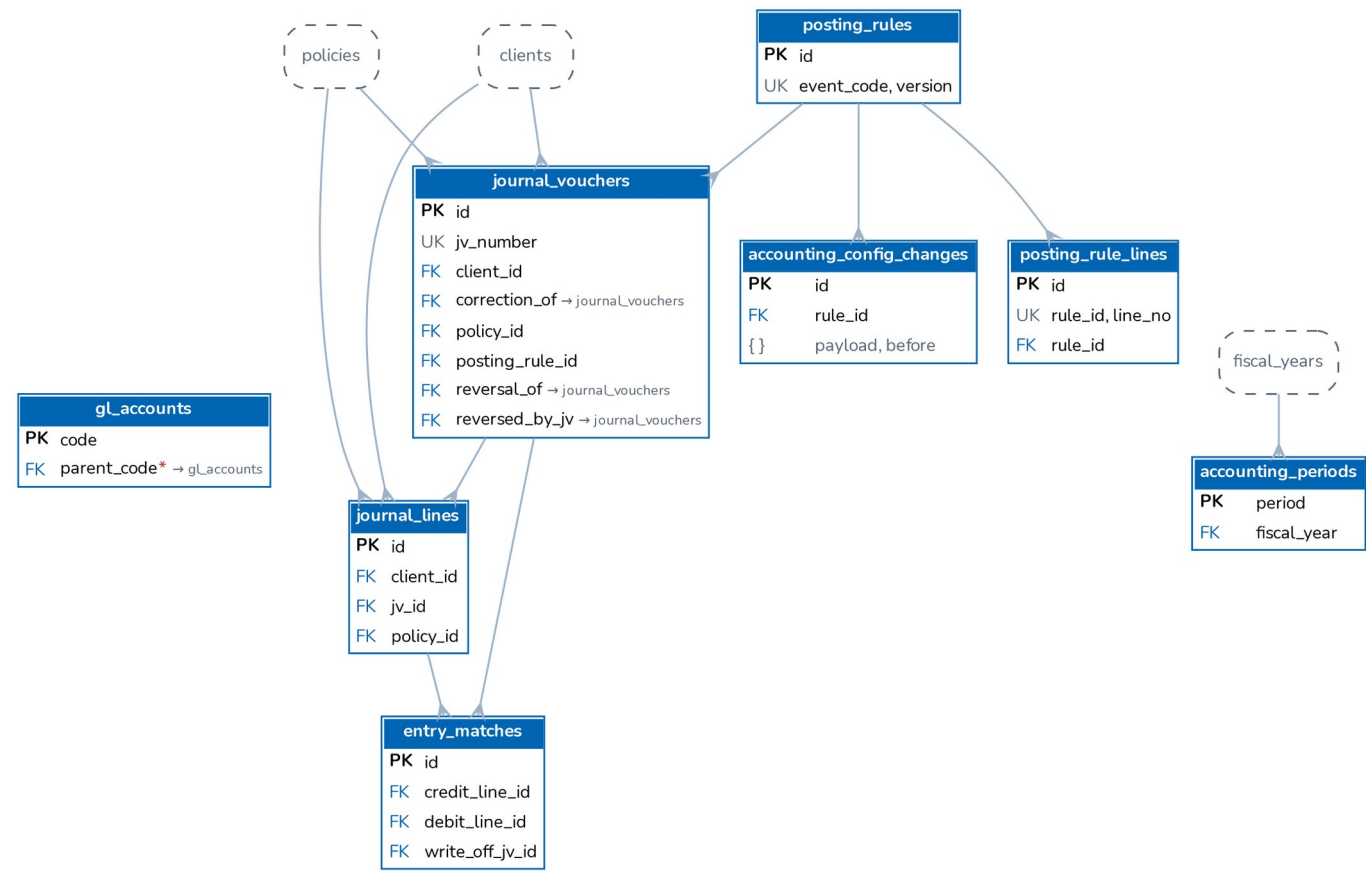


Figure 5. General ledger: chart of accounts, journals, periods, matching and posting rules

Journal vouchers hold balanced journal lines coded to the chart of accounts (**g_l_accounts**). Open-item accounts are reconciled by **entry_matches** (debit line to credit line). Journals point back to their source document through **journal_vouchers.reference_type** and **reference_id**.

4.6 Period end and tax

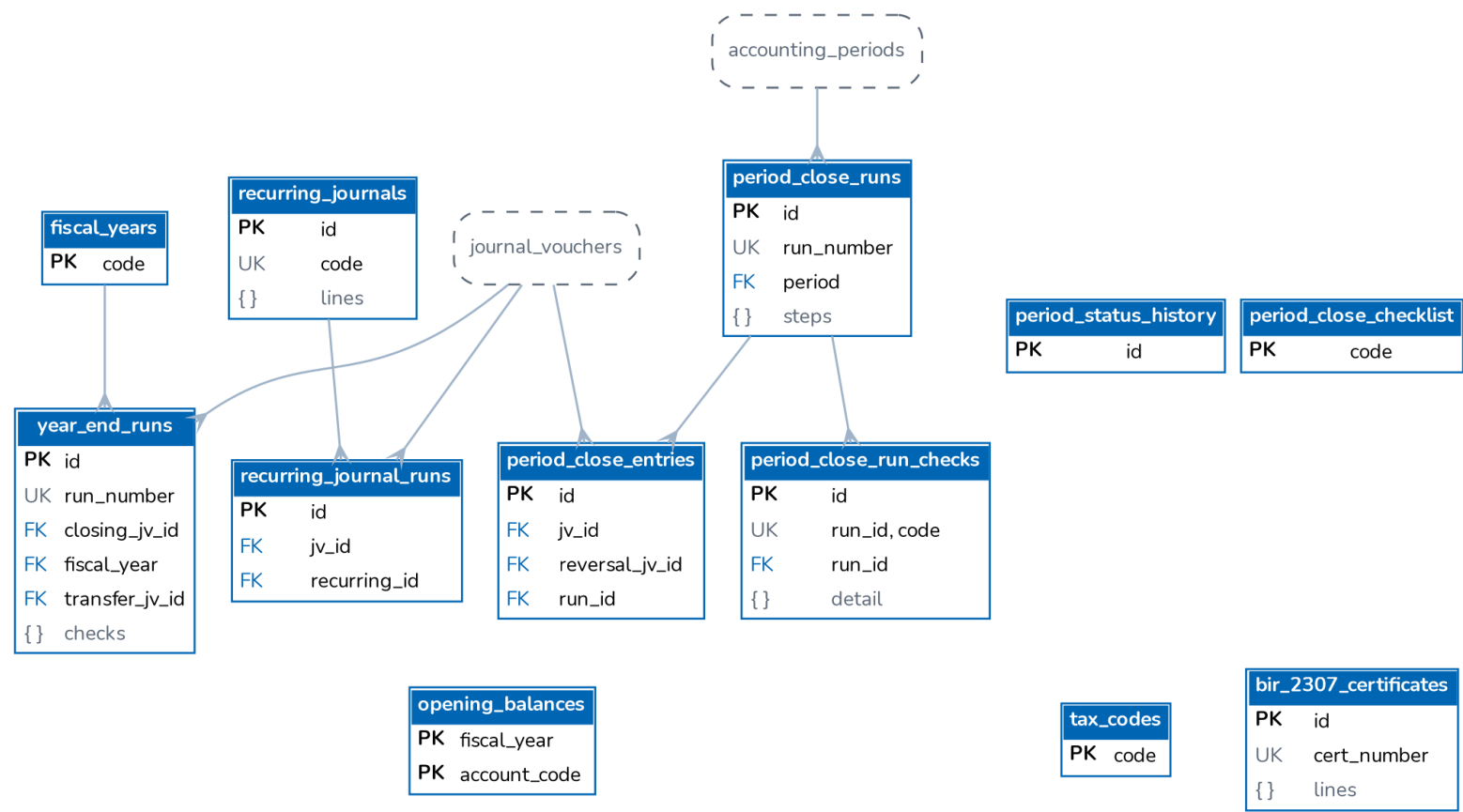


Figure 6. Period end: fiscal years, close runs, recurring journals, year-end, tax codes and BIR 2307

4.7 Bank reconciliation

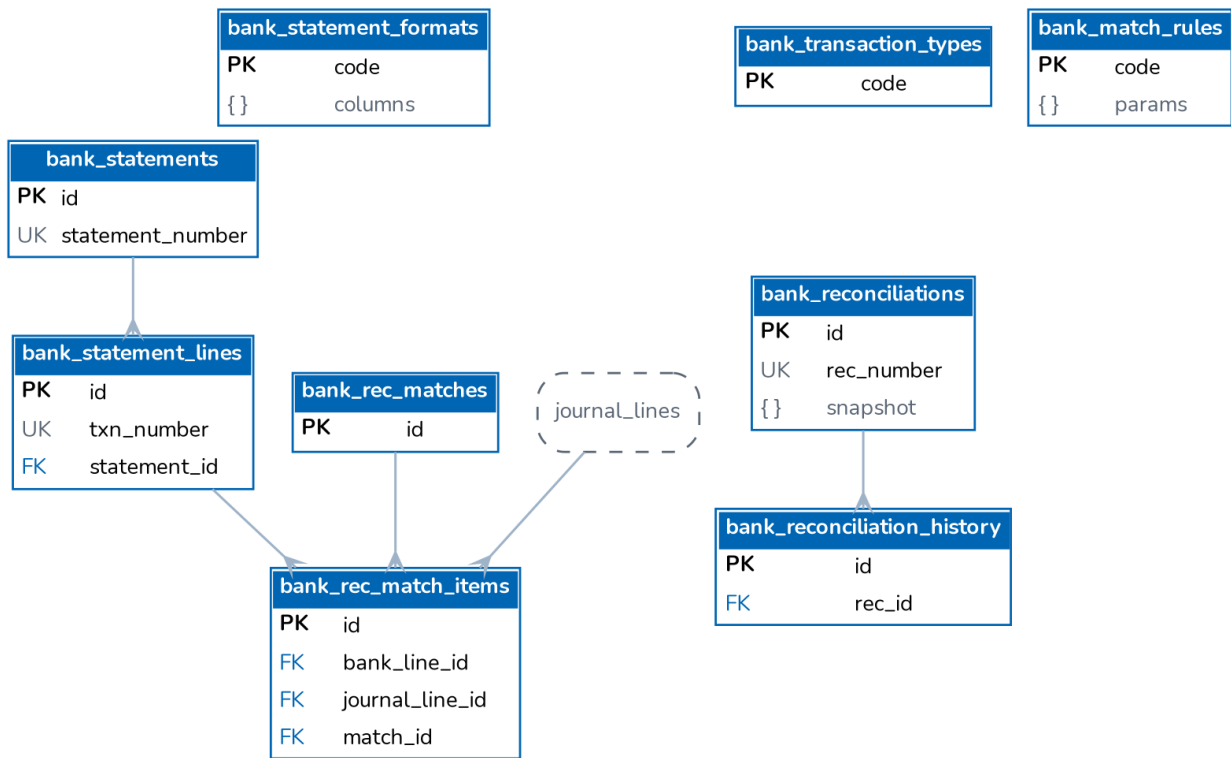


Figure 7. Bank reconciliation: statement formats, statements and lines, matches, reconciliations

A bank statement is imported with a statement format into `bank_statements` and `bank_statement_lines` (a line hash refuses duplicates). Matches link bank lines to cash-account journal lines (`bank_rec_match_items` with a *side*). A monthly reconciliation (`bank_reconciliations`) stores the statement figures and, once approved, locks the matches cleared up to the period end. The bank account itself is a master record linked to a GL cash account.

4.8 Disbursements and petty cash

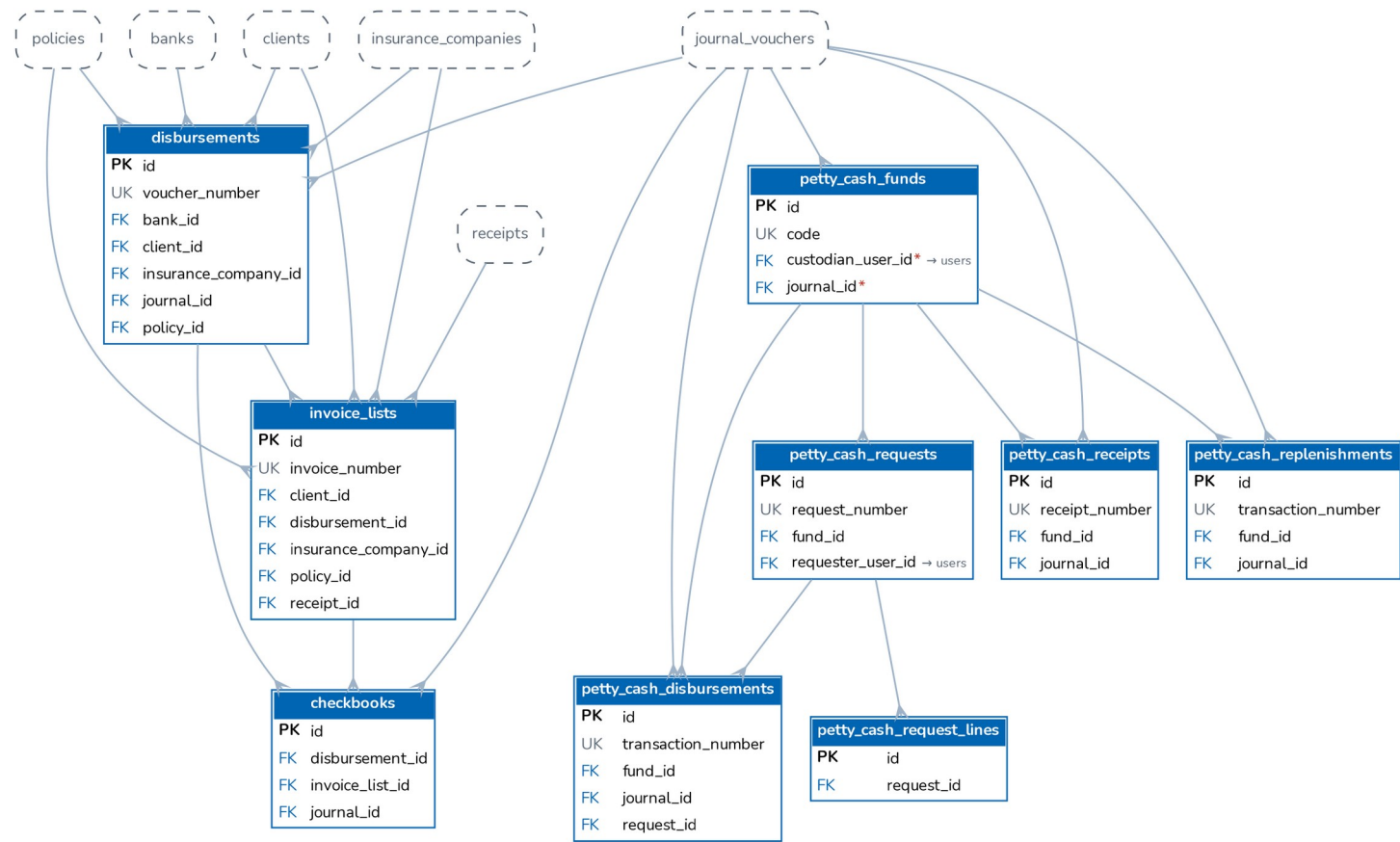


Figure 8. Disbursements (payment vouchers, invoice lists, cheques) and petty cash

4.9 Commission and incentive

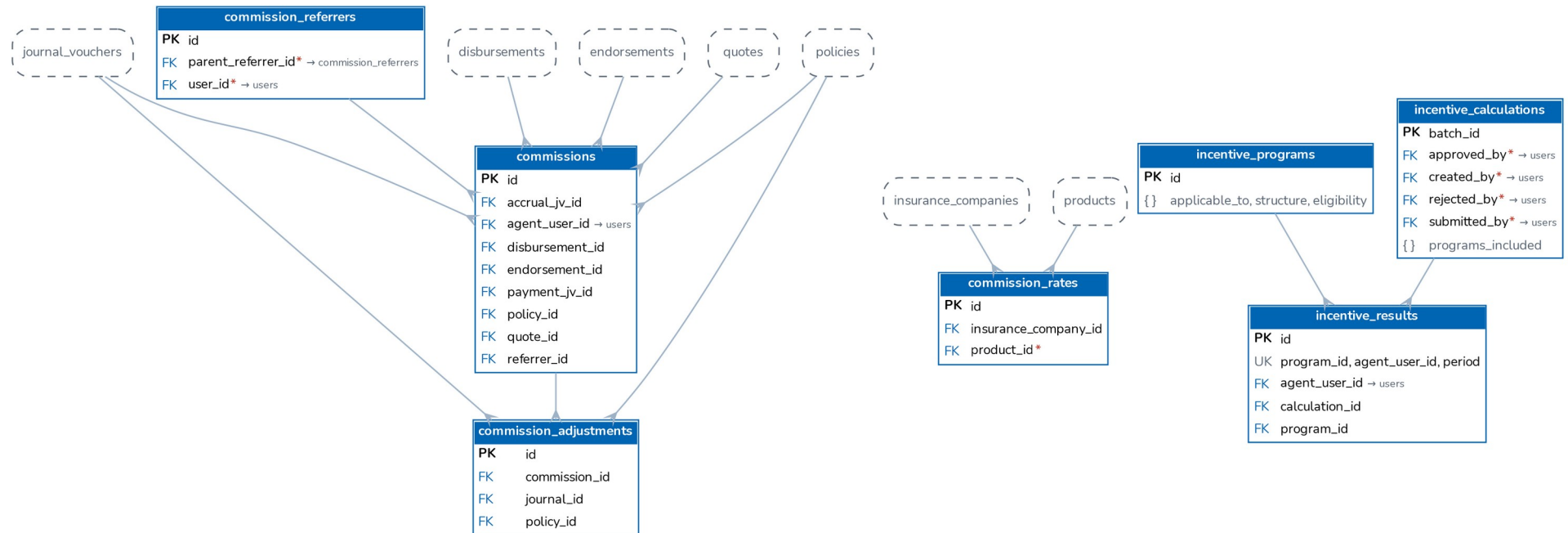


Figure 9. Commission lines, adjustments, the rate matrix, referrers and incentive programmes

Commission lines are created per policy, endorsement and referrer in the referral chain (`commissions_policy_referrer_uq`), link to the accrual and payment journals and to the payment voucher that paid them. `commission_adjustments` records the reversal or clawback when premium is returned. The rate of a new policy comes from `commission_rates` (insurer, product, line, policy type, effective dates), else the insurer's default rate. Referrers form a hierarchy (`parent_referrer_id`); they do not sign in.

4.10 Claims

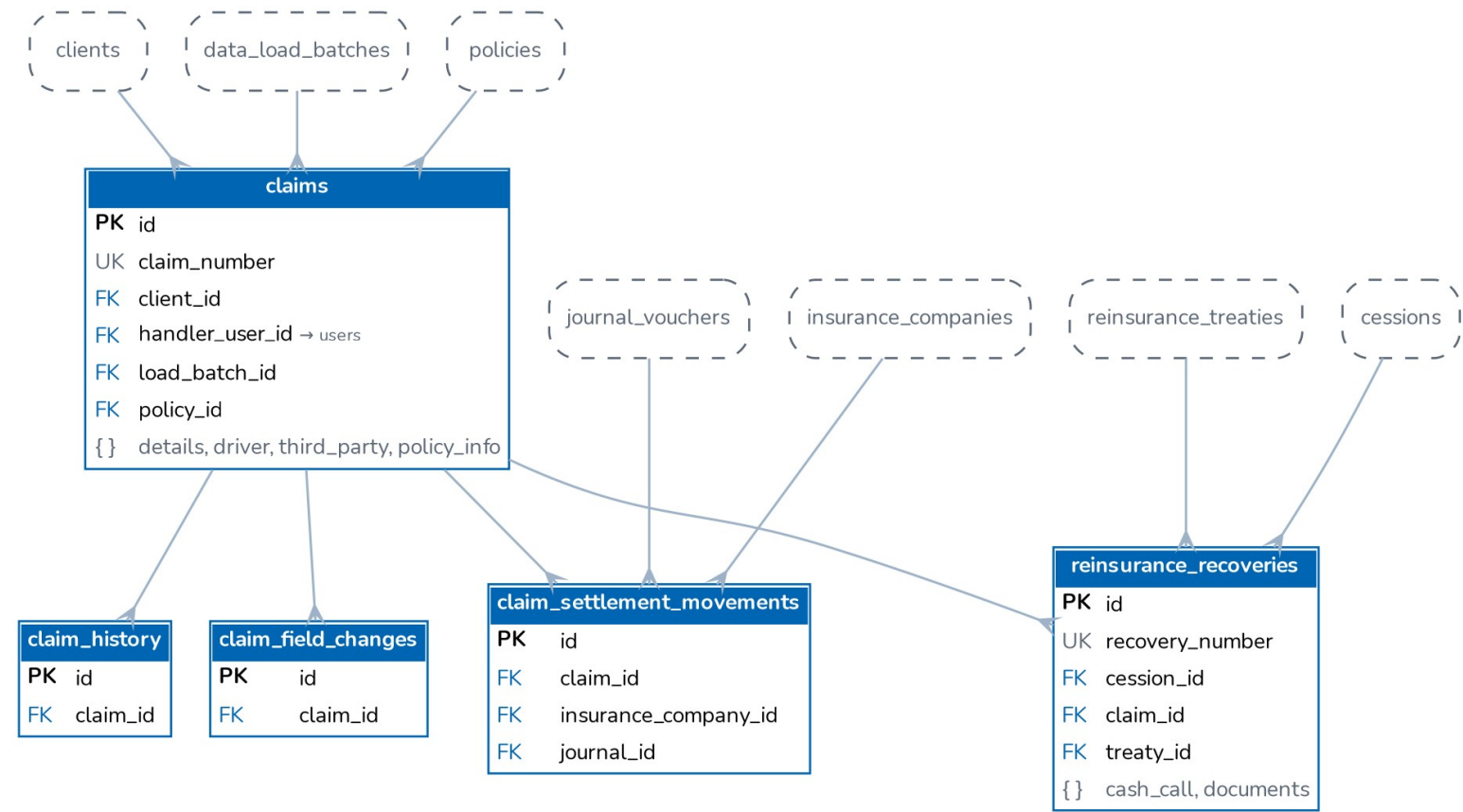


Figure 10. Claims, claim history, field-level changes, settlement cash and reinsurance recoveries

4.11 Remittance and direct bill

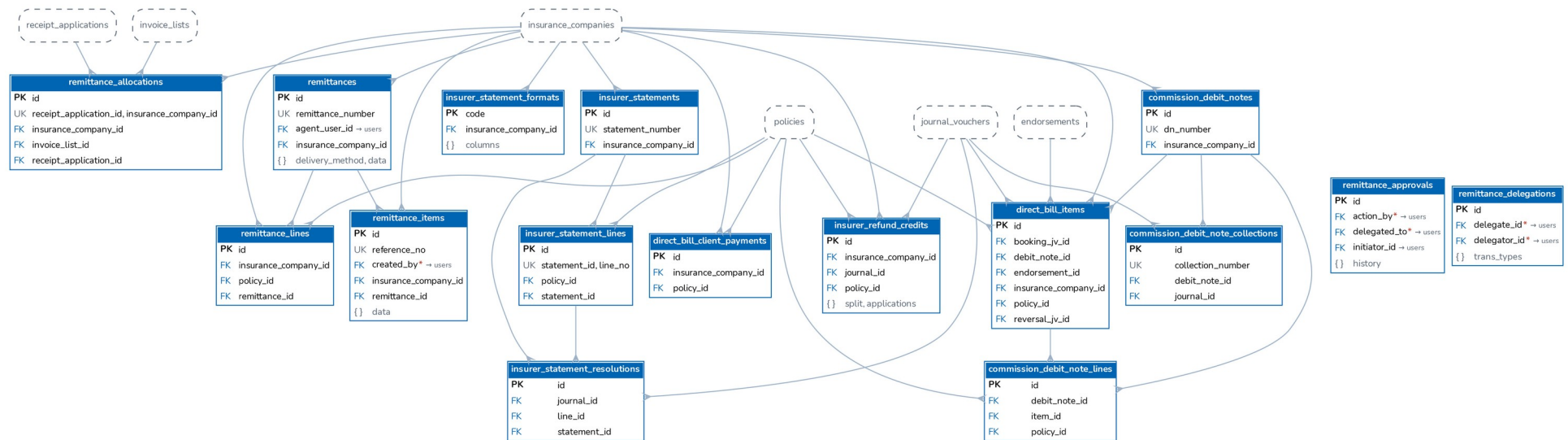


Figure 11. Remittances to insurers, co-insurance allocations, insurer refunds and direct-bill debit notes

Broker-billed premium is remitted to insurers through **remittances** with **remittance_lines** per policy; settlements, adjustments and exceptions are **remittance_items**; **remittance_approvals** is a multi-level approval register (entity, levels, SLA, history). For a co-insured policy **remittance_allocations** splits each collection by insurer share. **insurer_refund_credits** holds refunds due from an insurer on returned premium already remitted; they are netted against the next remittance. Direct-billed policies produce **direct_bill_items** (commission receivable) billed to the insurer on **commission_debit_notes** and settled by **commission_debit_note_collections**.

4.12 Reinsurance

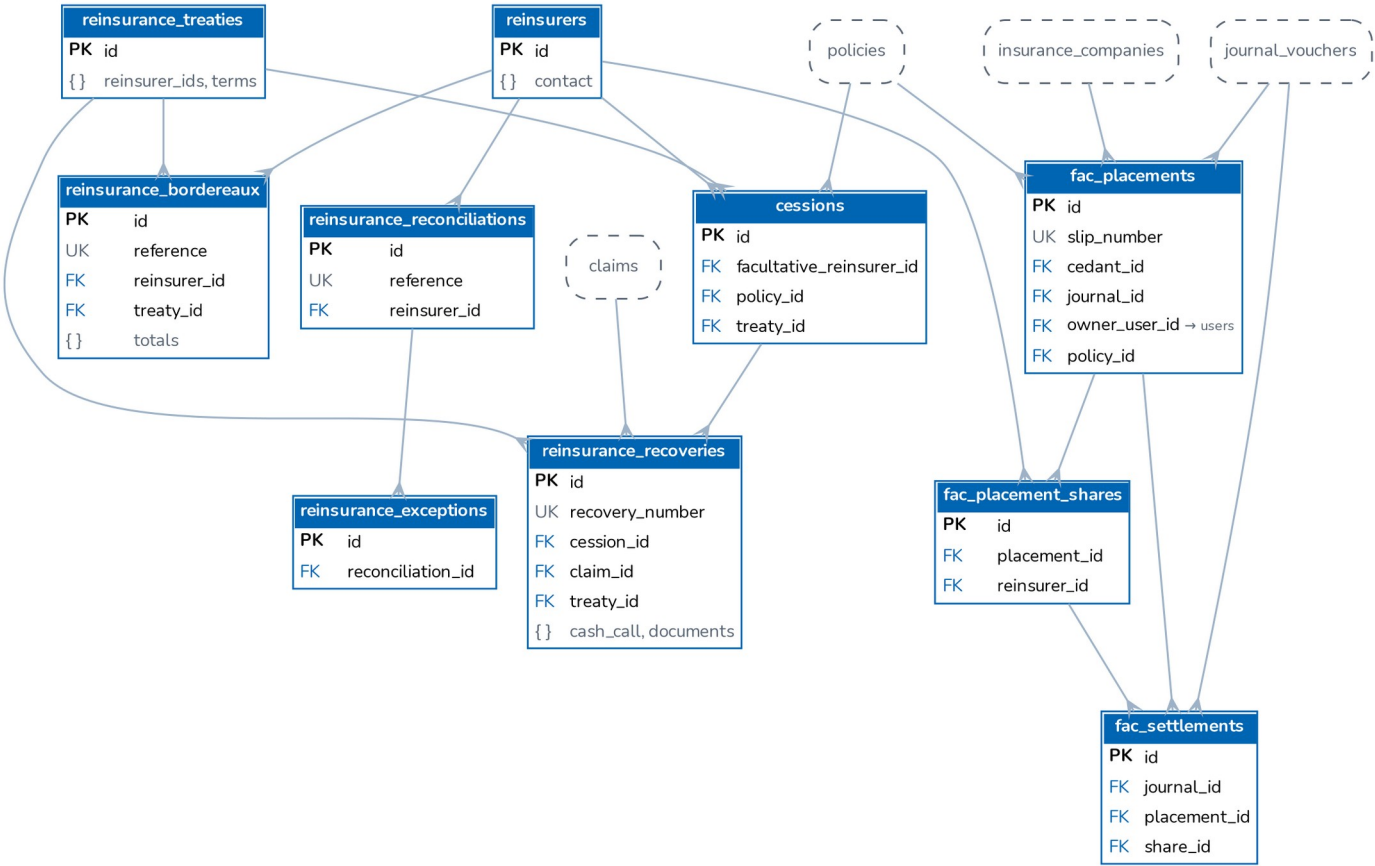


Figure 12. Reinsurers, treaties, sessions, recoveries, bordereaux and reconciliations

4.13 Configuration, security and platform

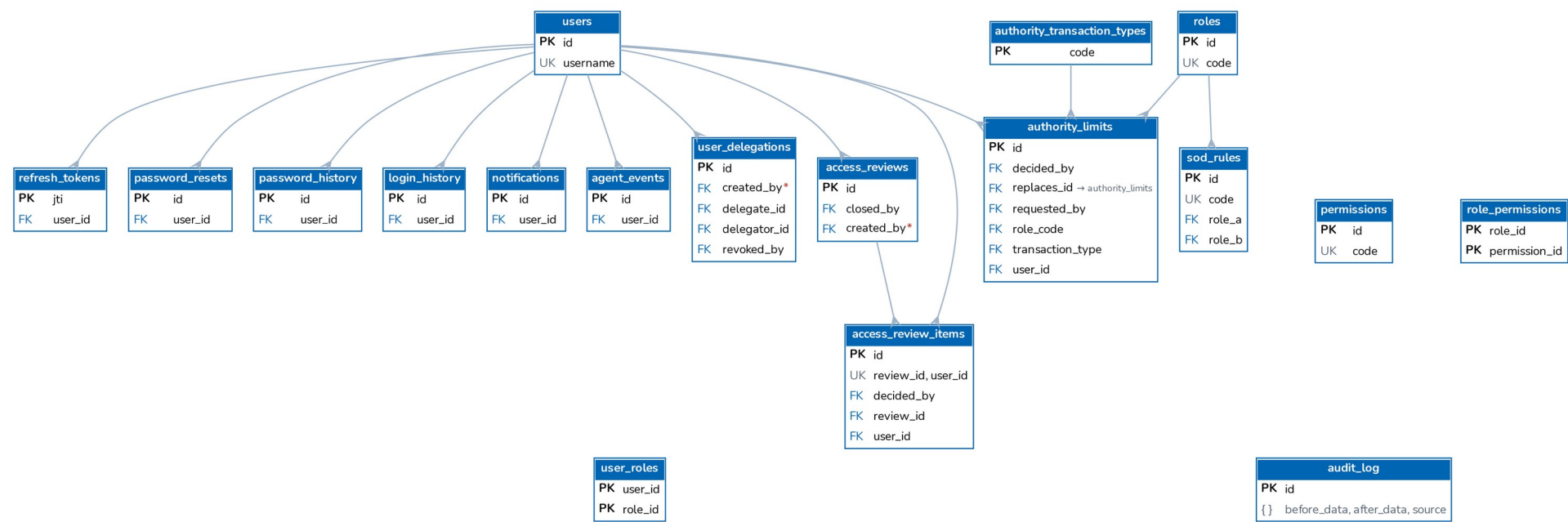


Figure 13. Security and identity tables

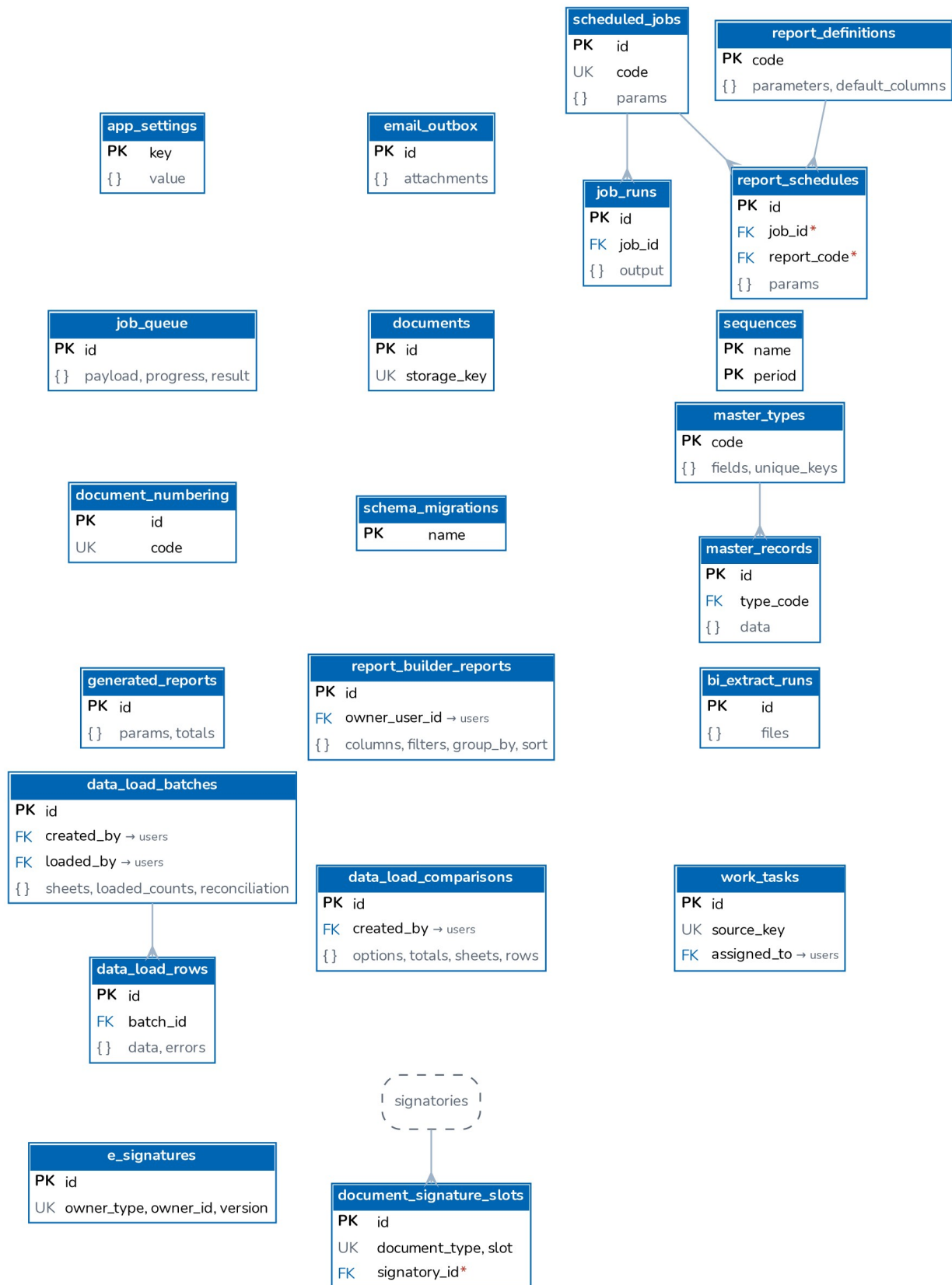


Figure 14. Configuration and platform services tables

4.14 Reference masters and product configurator

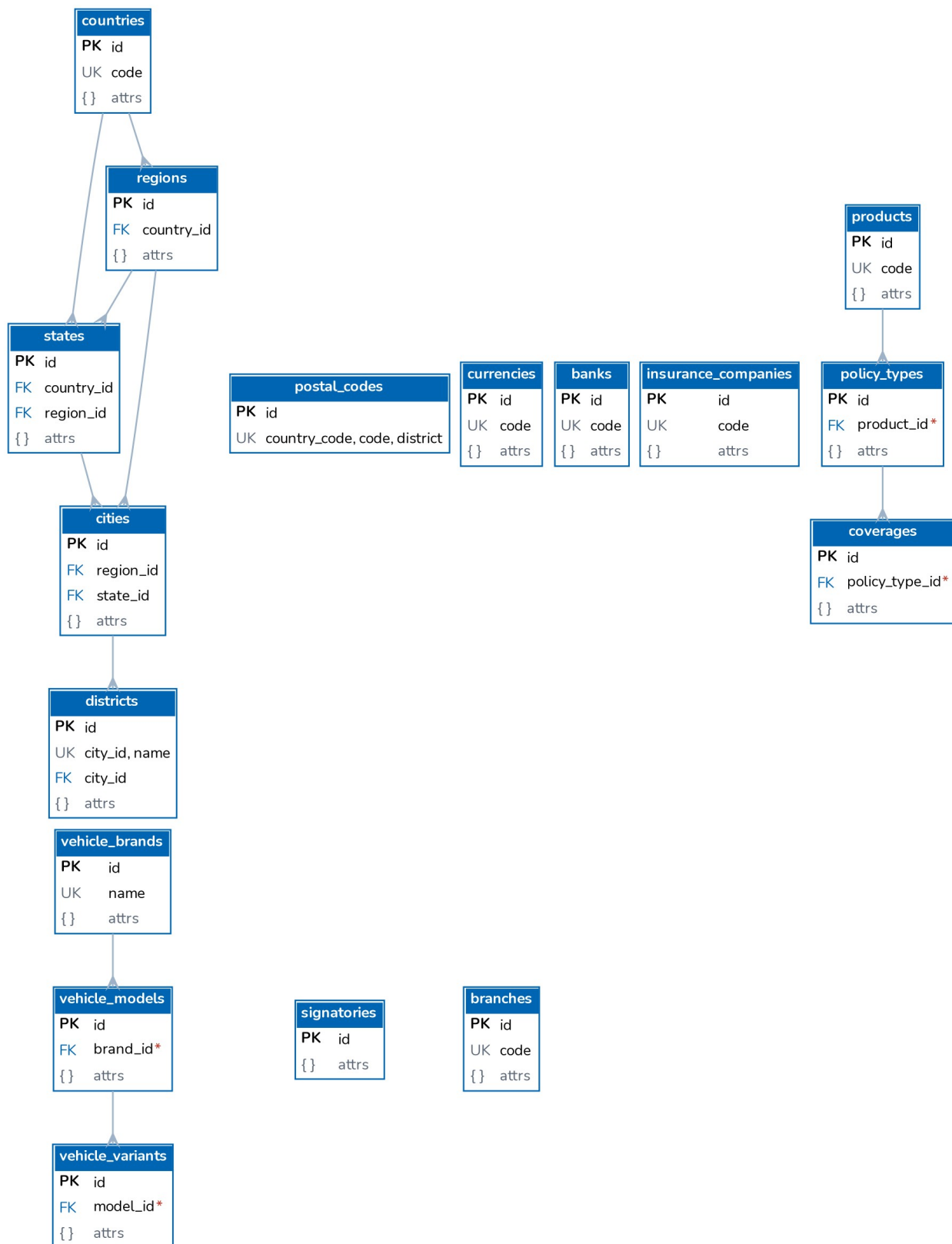


Figure 15. Reference masters

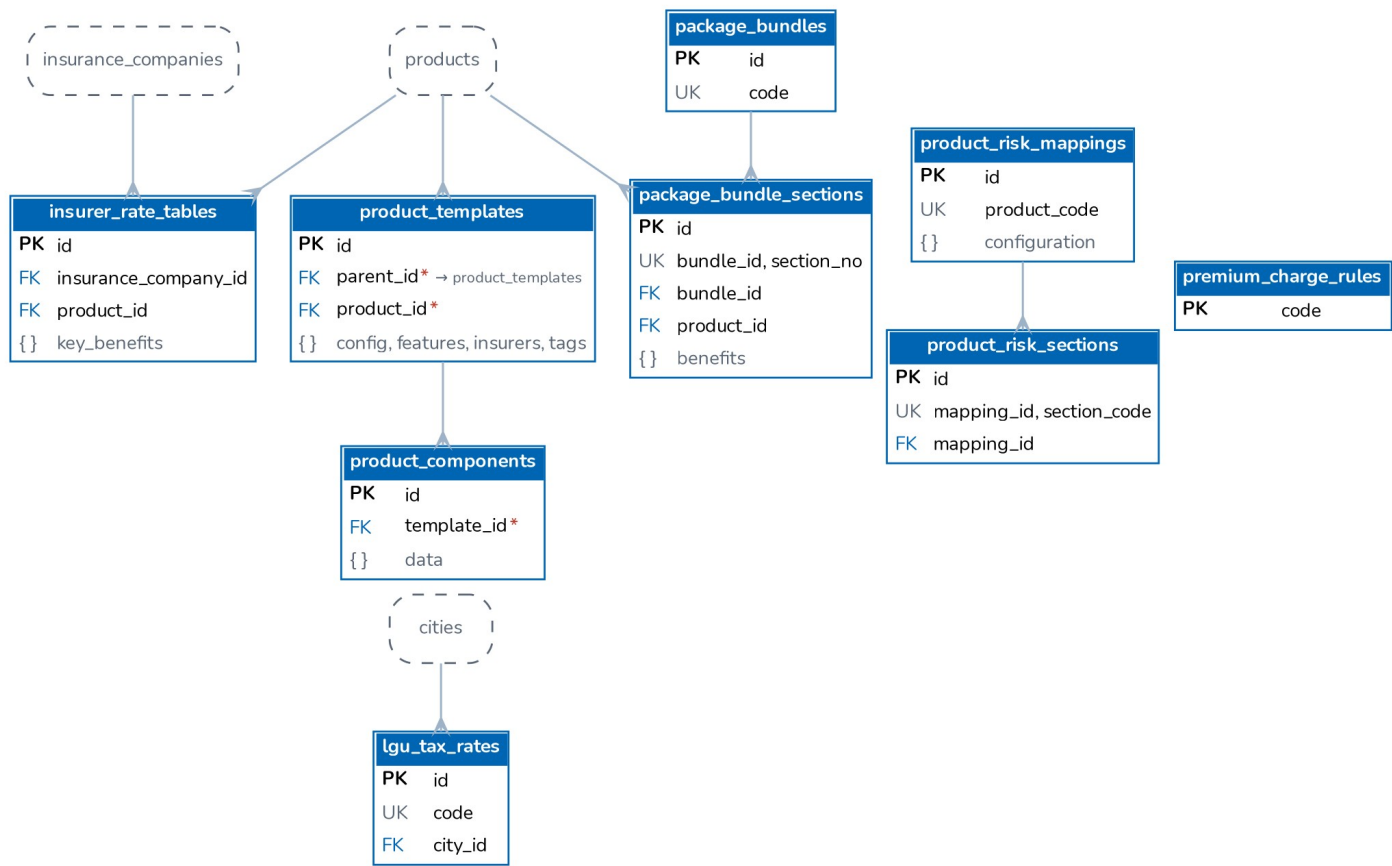


Figure 16. Product configurator and pricing

4.15 Distribution and marketing

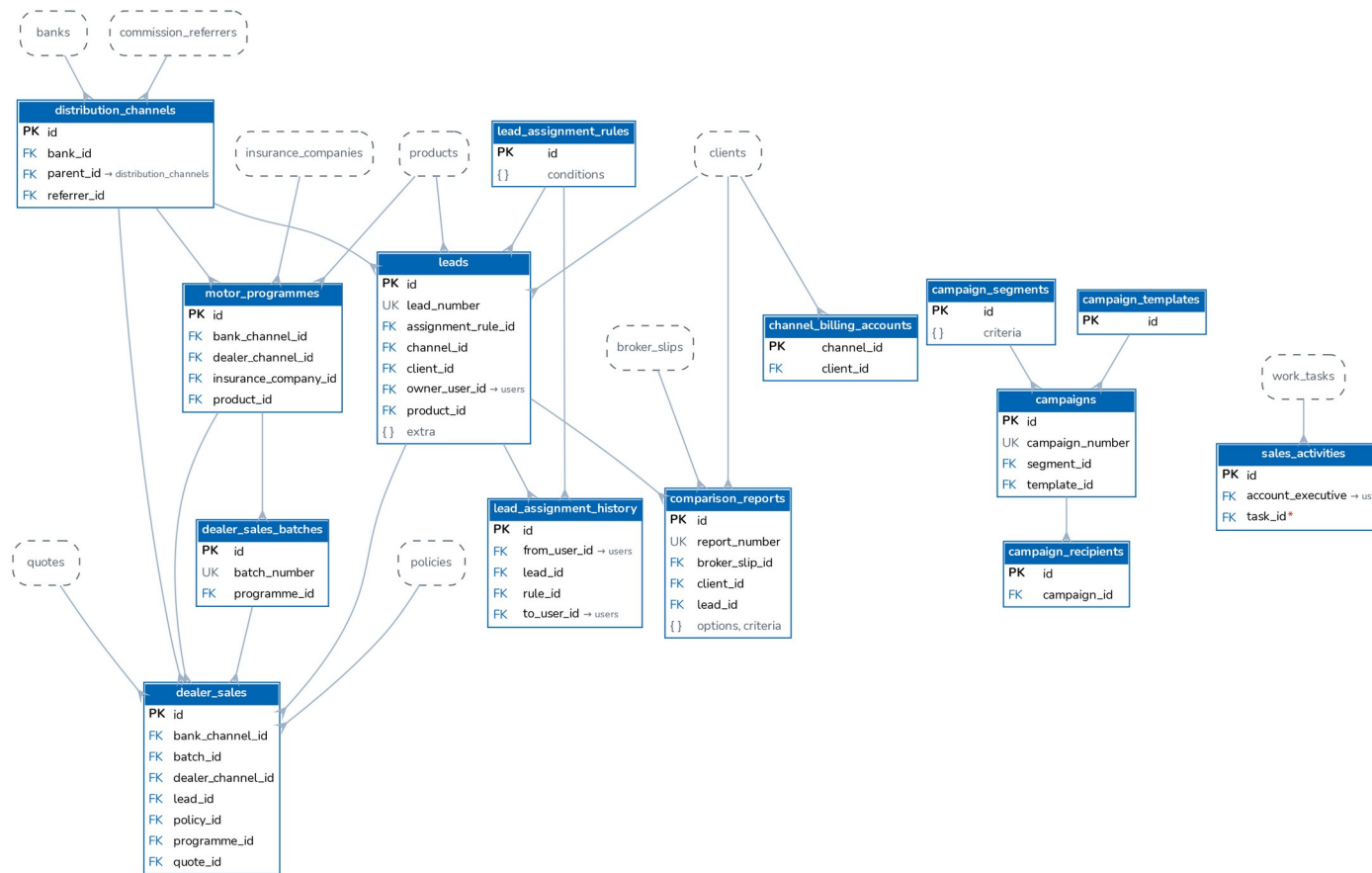


Figure 17. Distribution and marketing: lead assignment, channels, dealer programmes, comparison reports, campaigns, sales activities

Lead assignment rules route a new prospect; every assignment is kept in `lead_assignment_history`. `distribution_channels` is one hierarchy of dealer groups and branches, financing banks and affinity partners (`parent_id`); dealer programmes (`motor_programmes`) set the terms of the cars a dealer sells and uploaded `dealer_sales` become quotations or policies. Campaigns send to a segment through a template; `campaign_recipients` records consent, result and opt-out. `sales_activities` (package G) logs calls and visits and creates follow-up tasks in `work_tasks`.

4.16 Fleet, marine open cover, claim documents and motor repairs

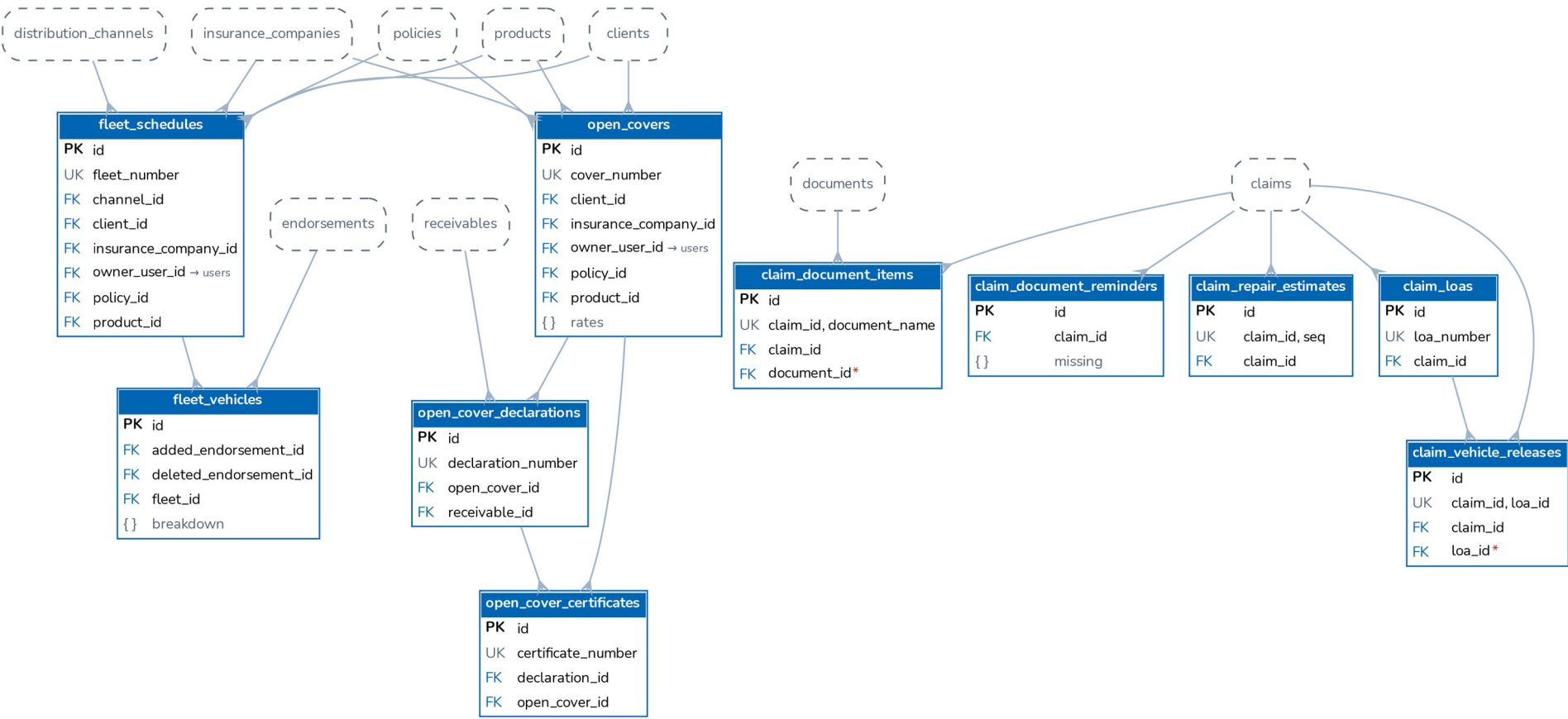


Figure 18. Fleet schedules, marine open covers, claim document checklist and motor repair records

4.17 Payables, fixed assets, BIR and override commission

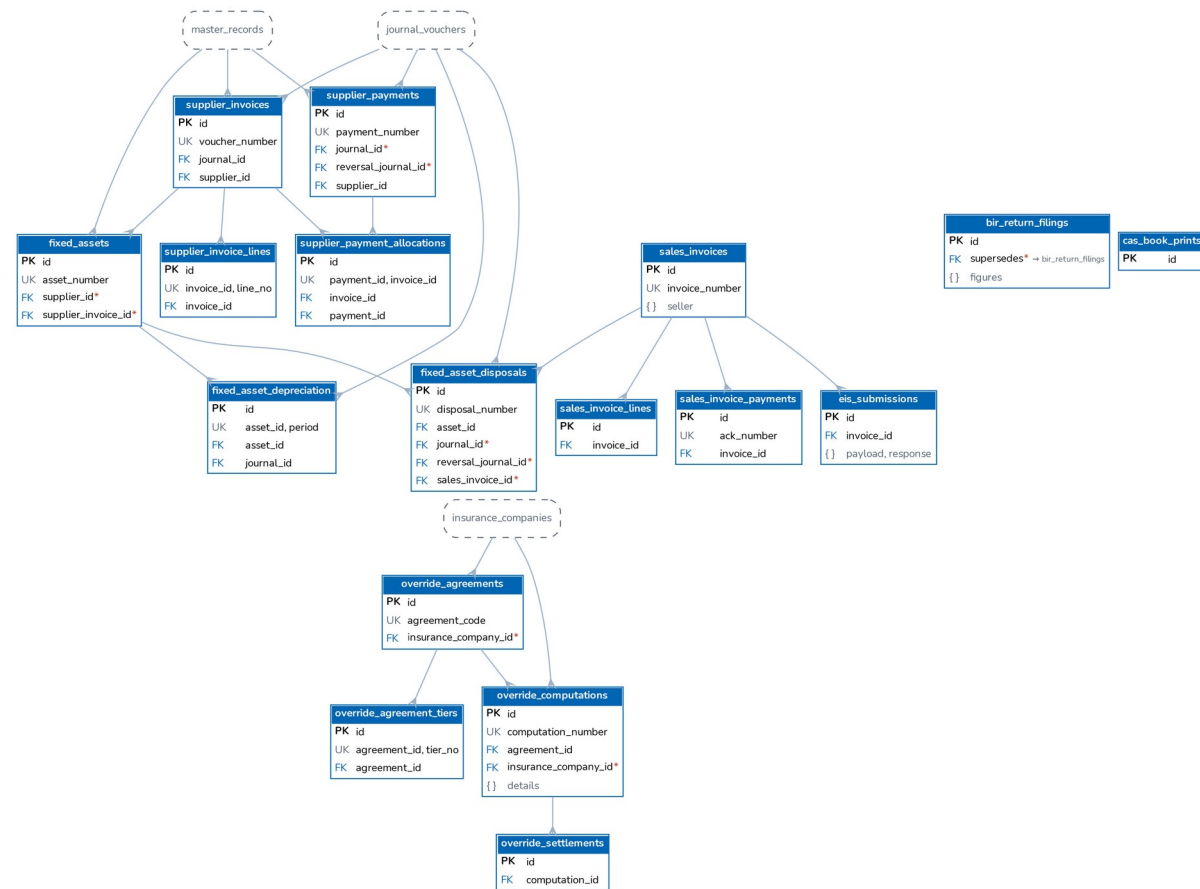


Figure 19. Accounts payable, fixed assets, BIR return filings, sales invoices and EIS, override commission

Supplier invoices post on approval (`ap.invoice`) and payments through `supplier_payment_allocations` (`ap.payment`); assets depreciate monthly (`fa.depreciation`) and are disposed of with a gain or loss (`fa.disposal`, package G). The BIR returns are computed from the ledger and recorded in `bir_return_filings`; every sales invoice is queued in `eis_submissions`. Override commission agreements, computations and settlements accrue and settle insurer overriding commission.

4.18 Compliance

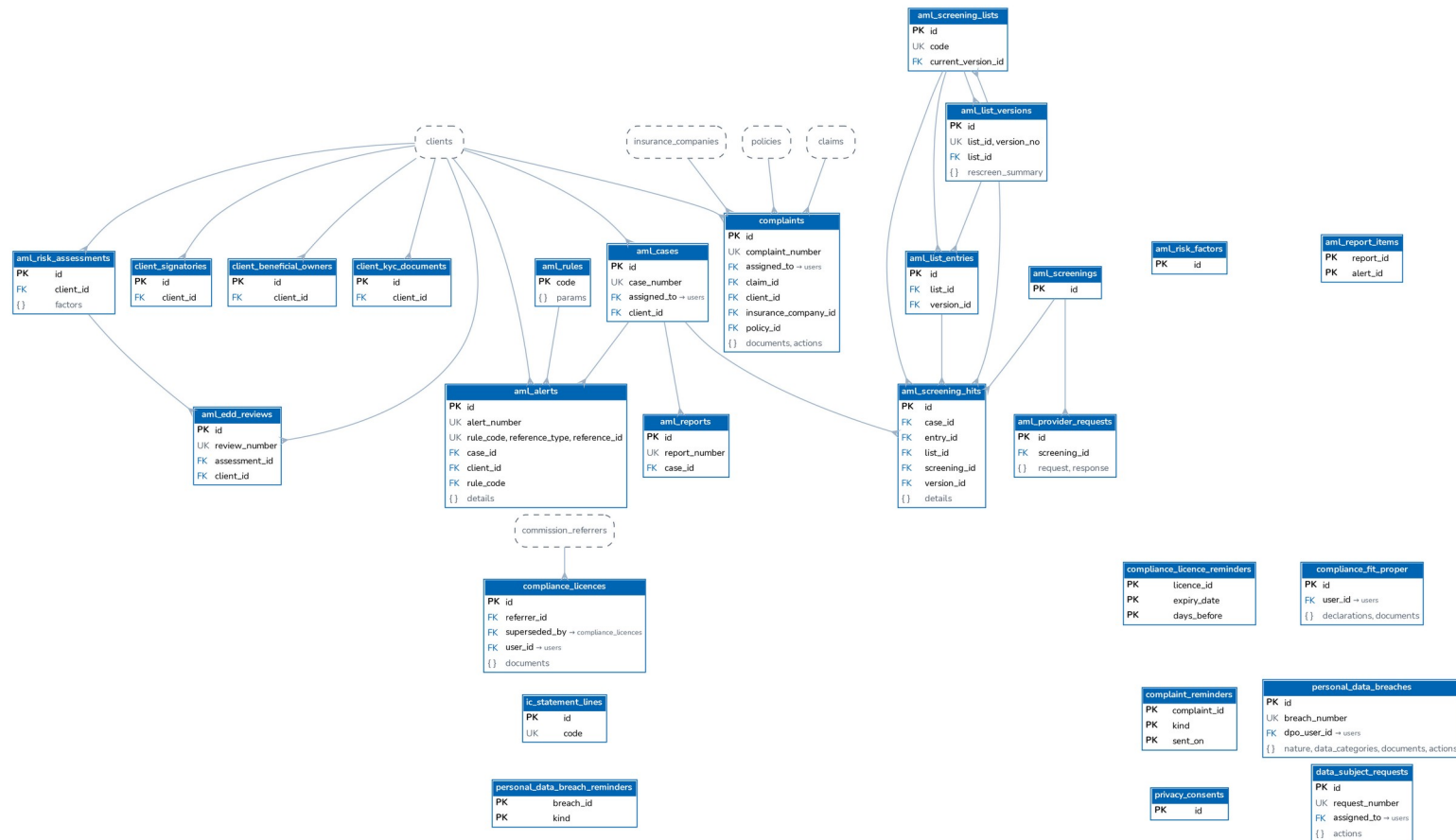


Figure 20. AML/CFT, Insurance Commission registers, complaints, breaches and data privacy

The AML tables hang off **clients**: KYC records of juridical clients, risk assessments and EDD reviews, screenings and their hits against the screening lists, alerts raised by monitoring rules, cases and AMLC report files. Package B adds the licence register, fit and proper records, the IC statement mapping, the complaints register and the breach register. With package B, TIN, ID and bank account numbers are encrypted in the database by the triggers `pii_protect_columns` and `pii_protect_json` with blind indexes (`*_bidx`) for exact search (document 05).

4.19 Integrations

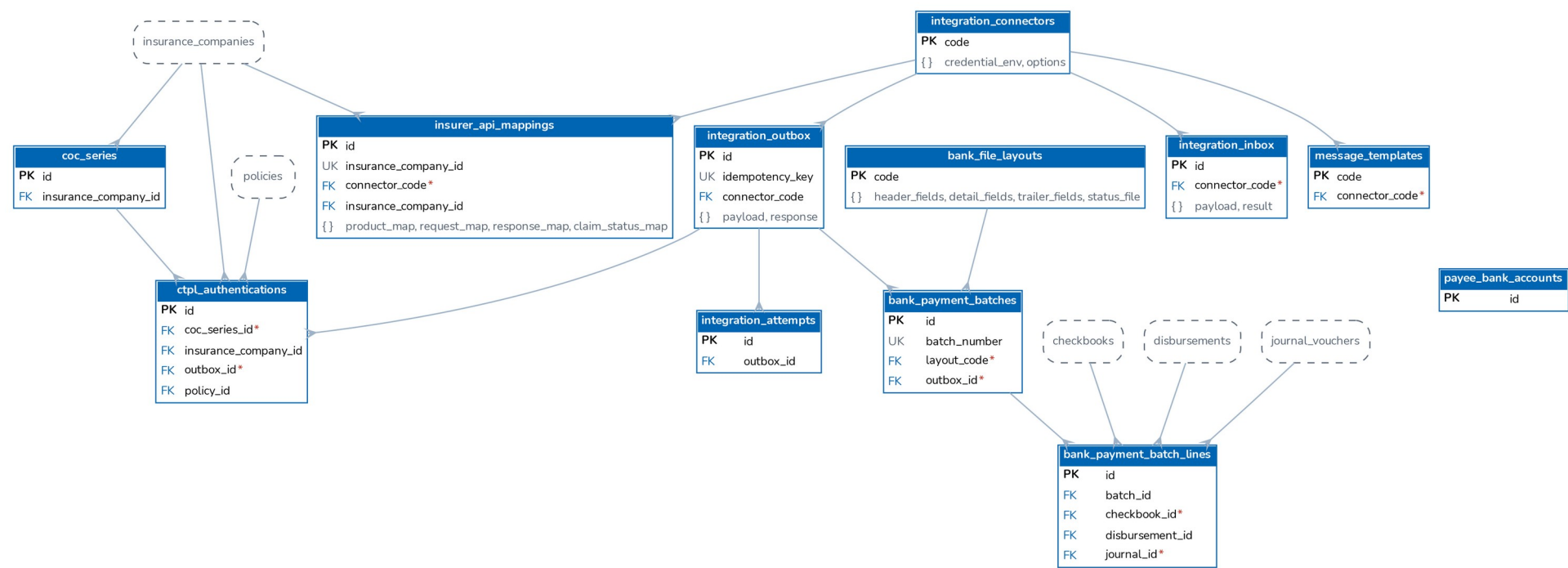


Figure 21. Integration framework: connectors, outbox, attempts, inbox, messaging, CTPL, insurer mappings and bank payment files

Every message to a third party is a row of `integration_outbox` written in the business transaction; `integration_attempts` keeps each try and `integration_inbox` each message received. Connectors keep only the names of the environment variables that hold their credentials.

5 Migrations

Aspect	Implementation (<code>backend/src/db/migrate.js</code>)
Format	Plain SQL files <code>NNNN_name.sql</code> in <code>backend/src/db/migrations</code> , applied in file-name order: 0001-0004 core, masters, operations, finance; 0010-0021 sales, claims, renewals; 0030-0034 ledger, receipts, disbursements, commission, petty cash; 0040-0050 configuration, product configurator, remittance, reinsurance, incentive, reports; 0060-0097 security and later changes; 0110-0113 branding and the Company letterhead; 0115-0117 role inheritance and bank reconciliation; 0120-0127 co-insurance, document numbering, commission rates, insurer credit terms, placement; 0130-0136 posting rules, period end, tax codes; 0140-0152 broker roles, business-date defaults, foreign-key indexes, accounting follow-ups, go-live imports and the e-mail sender.
Tracking	Table <code>schema_migrations(name, applied_at)</code> ; a file is applied once.
Execution	On every API start (<code>server.js</code>), before the seed. The migrate step first takes the advisory lock <code>hashtext('brokerverse.migrate'), hashtext('schema_migrations')</code> ; instances that start together wait (up to <code>MIGRATION_LOCK_TIMEOUT_MS</code> , 10 minutes) and then find nothing to do. Each pending file runs in its own transaction with its <code>schema_migrations</code> row; on error the transaction is rolled back and the API does not start. <code>npm run migrate</code> runs the same code; <code>npm run db:reset</code> drops and recreates the schema (development only).
Idempotency	Files use <code>CREATE TABLE IF NOT EXISTS, ADD COLUMN IF NOT EXISTS, CREATE INDEX IF NOT EXISTS, ON CONFLICT DO NOTHING</code> and guarded updates, so they are safe on a database that already holds data. An applied file is never edited; fixes go forward in a new file.
Readiness	<code>GET /api/health</code> answers 503 until no migration is pending, so a load balancer does not route to an instance with an older schema.
Direction	Forward only. Migrations only add; rolling back data means restoring a snapshot (document 09).

Recommended / to be confirmed by the business and DevOps: Keep destructive changes (column drops, type changes) in a separate release after the code that no longer uses them is live. Start one instance first when a release contains migrations (`deploy/REFERENCE.md`, Upgrading).

6 Seed data

`backend/src/db/seed.js` runs on every start after the migrations and with `npm run seed`. Every step is idempotent: rows are inserted by natural key or fixed id, so administrator edits are kept.

1. In code: the roles, permissions and grants, and the administrator **BrokerVerse** (password from `ADMIN_PASSWORD`; without it a random password is generated, logged once and must be changed at the first sign-in).
2. `seeds/settings.json` (configuration keys, inserted only if missing) and `seeds/jobs.json` (the scheduled jobs; the period-end and bank-matching jobs come from migrations).
3. Encryption of any two-factor secret still stored in clear (upgrade path).
4. The SQL files `seeds/*.sql` (**reference data**, always) and `seeds/sample/*.sql` (**sample data**, only when `SEED_SAMPLE_DATA` is on), interleaved by file name so a sample file runs after the reference file it depends on.

Class	Content
Reference	Settings (659 keys in 64 groups); 8 roles and 98 permissions; Philippine geography from the PSGC of the 2nd quarter 2026 (regions, provinces, cities and municipalities with ZIP codes; barangays as an optional load), banks, ID types, salutations, holidays and the IC insurer list (<code>seeds/12_ph_geography.sql</code> , <code>69_ph_practice_masters.sql</code>); currencies, banks, insurers, products and policy types; vehicle makes, models and variants; covers; head office branch; the Company master with the letterhead company; master screen registry and reference master records; chart of accounts (129 accounts in this database) with main and sub account masters; posting rules for 43 events and Account Determination; 85 document number series; tax codes (VAT, expanded withholding with ATC, DST, LGT); the month-end checklist; bank statement formats (three Philippine bank layouts and a generic one); product templates with the motor and CTPL tariff; report catalogue (40 reports); direct-bill and security configuration; BIR forms, operations and accounting rules, AML risk factors and rules, integration connectors in test mode (<code>seeds/72_</code> , <code>73_</code> , <code>75_</code>).
Sample	Fictional leads, clients, quotations, broker slips and placements, policies, endorsements, claims, renewals, bills, receipts, collections, disbursements, petty cash, commissions and referrers, remittances, reinsurance, incentives, journal vouchers, bank statements, demo branches, signatories and master records, and six sample users (five Sales & Marketing users and one Accounting approver).

`SEED_SAMPLE_DATA` accepts true or false (1 or 0, yes or no, on or off); unset, it is on in development and test and off with `NODE_ENV=production`. A database seeded with sample data is cleaned with `npm run purge:sample` (document 10).

7 Time zone, date and amount handling

Topic	Design
Storage	Instants in <code>timestamp_tz</code> (stored in UTC); calendar dates in <code>date</code> . The database time zone is Asia/Manila; the containers run in UTC.
Business date	<code>lib/dates.js</code> : "today" is the calendar date in <code>general.timezone</code> (seeded Asia/Manila), not the server's UTC date. <code>businessDate()</code> , <code>calendarPeriod()</code> , the report engine, document numbering and the column defaults (<code>numbering_business_date()</code> , migration 0145) use the configured zone, so between 00:00 and 08:00 Manila time the system never uses yesterday's date.
Job times	The scheduler reads cron expressions in <code>general.timezone: 0 6 * * *</code> runs at 06:00 Manila time.
Display	The front end formats dates with <code>general.date_format</code> (DD/MM/YYYY) and amounts with the configured currency and locale (en-PH grouping); the server uses <code>formatMoney()</code> and the PDF engine the same settings.
Money	<code>numeric</code> in the database, rounding by <code>round2()</code> ; taxes (VAT 12%, DST 12.5%, LGT 0.75%, expanded withholding by ATC) and commission rates come from settings, the tax codes and the rate matrix.

Note: Amounts are converted from `numeric` to JavaScript double-precision numbers. This is exact for amounts of up to about 90 trillion with two decimals, far above broking volumes, provided every arithmetic result is rounded with `round2()`, which the ledger and document code does.

8 Related documents

This document is one of eleven that together form the BrokerVerse OOTB Solution Architecture. They are maintained and issued together (source and build: [docs/architecture/README.md](#)).

No.	Document	File
01	Solution Component Diagram and Application Architecture	01_BrokerVerse_Solution_Component_Diagram_Application_Architecture.pdf
02	Database Design (this document)	02_BrokerVerse_Database_Design.pdf
03	Database Inventory	03_BrokerVerse_Database_Inventory.pdf
04	Technology Stack	04_BrokerVerse_Technology_Stack.pdf
05	Shared Service Components	05_BrokerVerse_Shared_Service_Components.pdf
06	Capacity and Performance	06_BrokerVerse_Capacity_and_Performance.pdf
07	High Availability and Resiliency	07_BrokerVerse_High_Availability_and_Resiliency.pdf
08	RTO and RPO	08_BrokerVerse_RTO_and_RPO.pdf
09	Backup and Recovery	09_BrokerVerse_Backup_and_Recovery.pdf
10	Data Archival, Housekeeping and Restoration	10_BrokerVerse_Data_Archival_Housekeeping_and_Restoration.pdf
11	Monitoring	11_BrokerVerse_Monitoring.pdf

9 Glossary

Terms and abbreviations used in this document.

Term	Meaning
Account determination	Settings accounting.account.<role> that give the GL account of each account role (cash in bank, premium receivable, due to insurer, commission income ...) used by the posting rules.
Advisory lock	PostgreSQL application-level lock (pg_try_advisory_lock, pg_advisory_lock) used so a job runs on one API instance at a time and only one instance applies migrations.
API	Application programming interface; here the BrokerVerse REST API under /api served by the Node.js backend.
ATC	Alphanumeric tax code of the BIR for a type of withholding, printed on BIR Form 2307.
Audit log / audit trail	Table audit_log: who changed which record, when, from which IP, with the before and after values.
BIR	Bureau of Internal Revenue (Philippines): tax authority whose rules drive record keeping and withholding tax certificates.
Bordereau	Periodic list of ceded premiums or claims sent to a reinsurer.
Broker slip	Request for quotation (BS-) that the broker sends to several insurers for one client risk; the answers are recorded as insurer offers.
Cession	The share of a policy's risk and premium passed to a reinsurer under a treaty or facultatively.
Co-insurance	Placement of one risk with several insurers, one of them the lead, each taking a share; premium, commission and claims are split by share.
Company master	Master record of the broker company; the primary company is the letterhead printed on every document and report.
Cron	Time-based schedule expression (minute hour day month weekday) used by the scheduled jobs.
CTPL	Compulsory Third Party Liability motor insurance (Philippines).
Debit note (DN)	Commission debit note raised by the broker to an insurer for direct-billed policies.
Direct bill	Billing mode in which the insurer collects the premium and the broker bills the insurer for its commission.
DST	Documentary Stamp Tax (Philippines), part of the gross premium.
Endorsement	Change to an issued policy (details, cover, premium or cancellation).
Fiscal year	Accounting year (FY2026) with twelve periods and an adjustment period 13 used by the year-end close.
FK	Foreign key: column that references the primary key of another table.
GIN	Generalised inverted index (PostgreSQL), used for JSONB searches.
Housekeeping	Daily job that deletes operational rows (job runs, sent e-mails, sign-in history, expired tokens, read notifications) after the retention days in System Settings.
IC	Insurance Commission (Philippines): regulator of insurers and brokers.
JSONB	PostgreSQL binary JSON column type, used for flexible document parts of records.
JV	Journal voucher: balanced set of debit and credit ledger lines.
KYC	Know your customer: identification data required before a policy is issued.
LGT	Local Government Tax (Philippines), part of the gross premium.
Maker-checker	Control in which a record created or submitted by one user must be approved by a different user.
Migration	Numbered SQL file in backend/src/db/migrations applied once, in order, on API start.
OR	Official receipt issued for premium received.

Term	Meaning
PK	Primary key: column(s) that uniquely identify a row.
Placement slip	Firm order (PS-) sent to the lead insurer and co-insurers; each participant binds before the policy is issued.
Posting rule	Versioned definition of the journal lines for one business event (policy issued, receipt applied ...); every system journal is built from it.
PV	Payment voucher (disbursement) to an insurer, referrer, client or supplier.
RDS	Amazon Relational Database Service, the managed PostgreSQL target.
Receivable / bill	Premium bill (INV-) raised when a policy, endorsement or renewal is issued.
Remittance	Payment of collected premium, net of commission, from the broker to the insurer.
RPO	Recovery point objective: the maximum acceptable data loss, measured in time.
RTO	Recovery time objective: the maximum acceptable time to restore service.
Seed	Idempotent data load run on every API start: reference data always, sample data only when SEED_SAMPLE_DATA is on.
SLA	Service level agreement / target time, e.g. for remittance approvals.
Treaty	Reinsurance contract under which risks are ceded automatically within agreed limits.
VAT	Value-added tax (12 % in the Philippines).
Year-end close	Run (YEC-) that posts the closing entries in period 13, carries the balances into the next year and locks the year.
zod	TypeScript-first schema validation library used to validate request bodies and parameters.